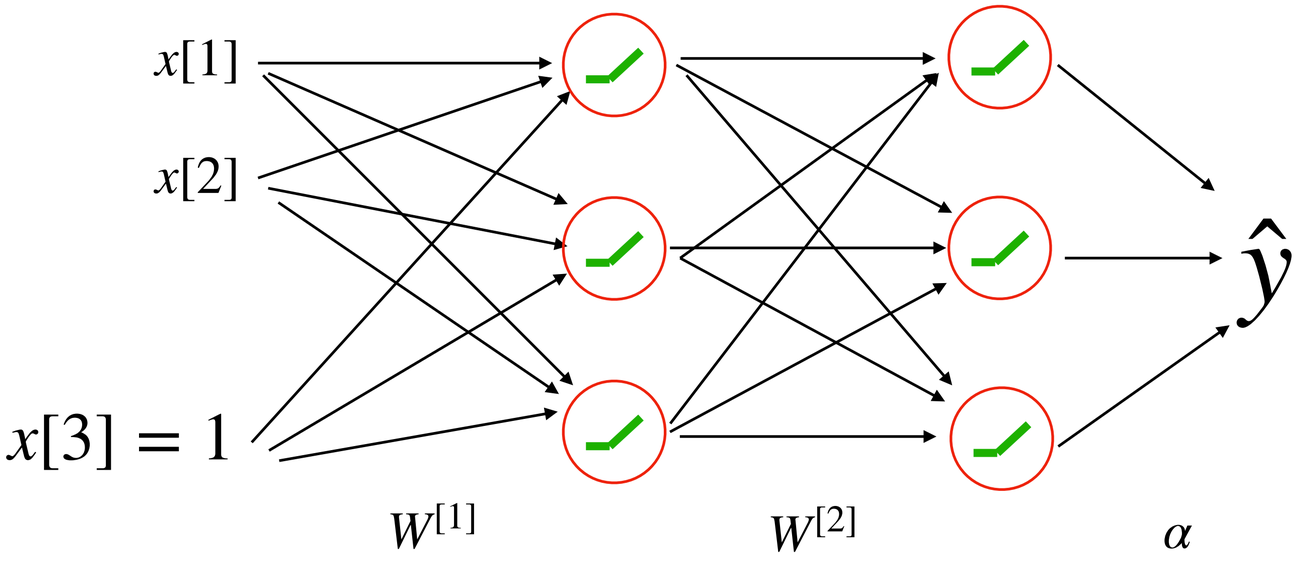
Neural Network:
Training & Backpropagation

Recap

A two layer fully connected feedforward NN:



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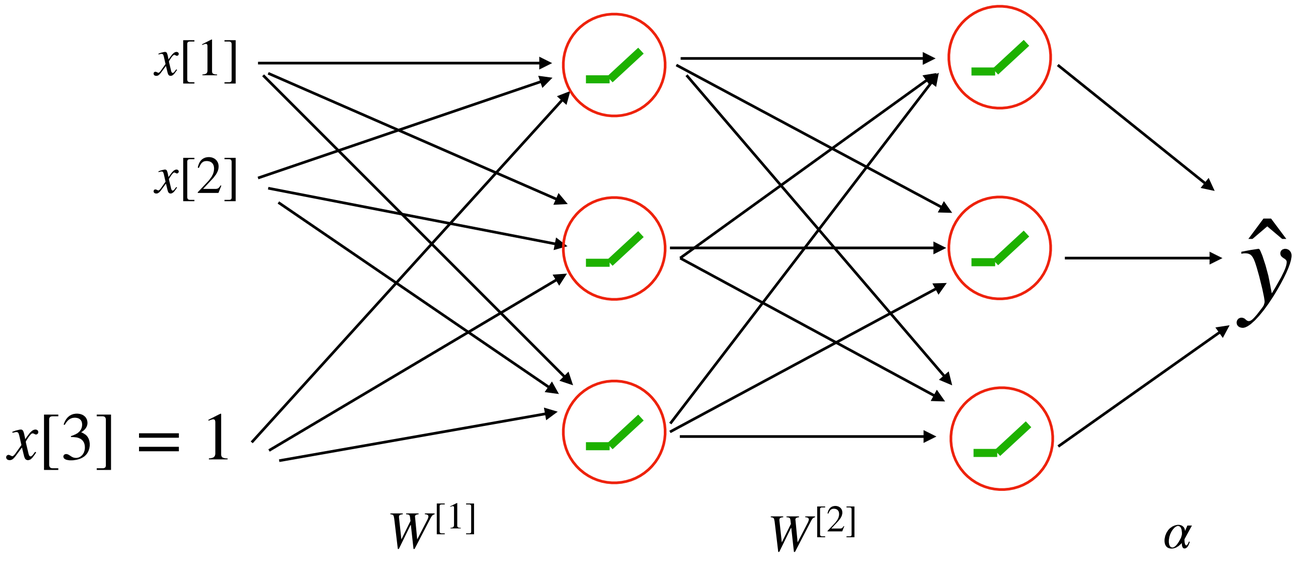

$$h(x) := \alpha^\top \mathrm{ReLU}\left(W^{[2]}\mathrm{ReLU}\left(W^{[1]}x\right)\right) + b$$

Outline of Today

1. Training NNs via SGD

2. A Naive approach of computing gradients

3. Backpropagation: efficient way of computing gradients

Training neural network via SGD

Square loss on training example (x, y)

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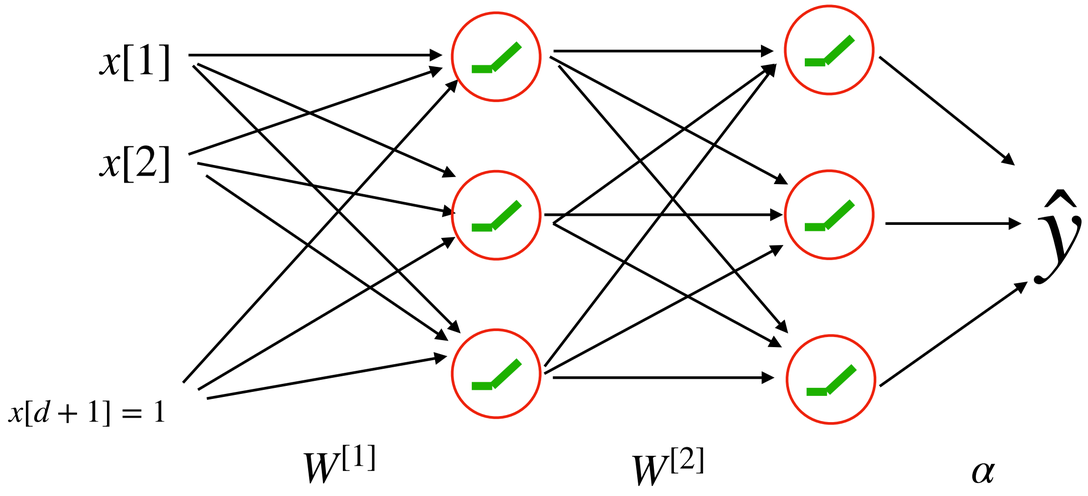

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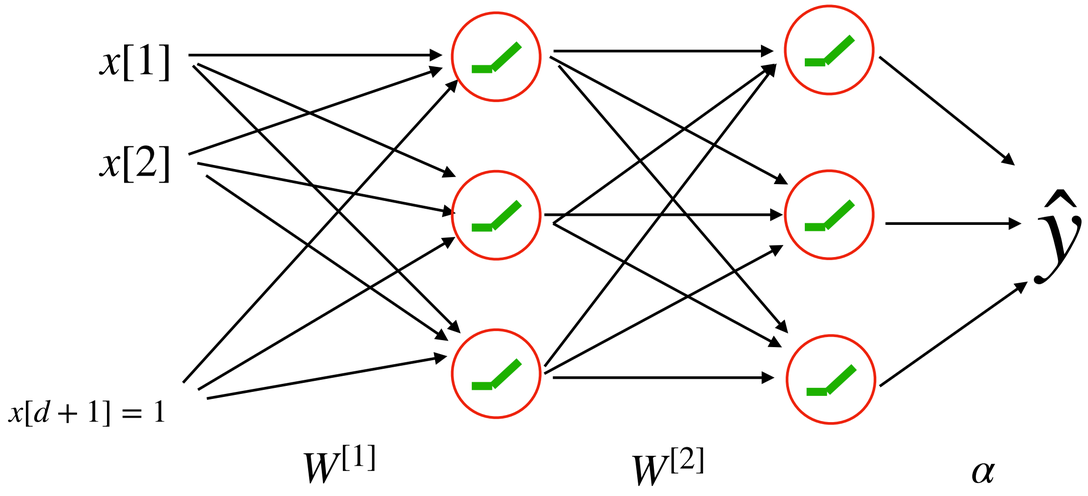

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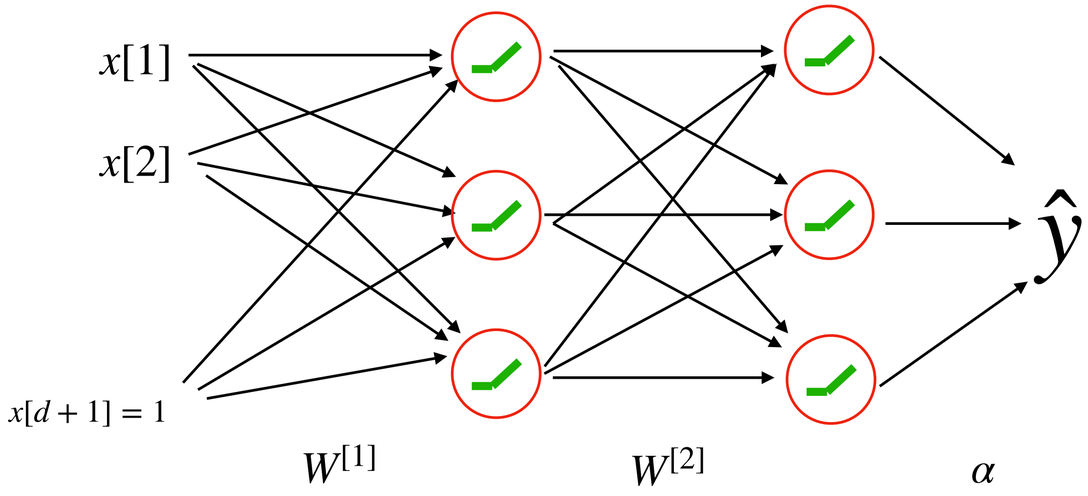

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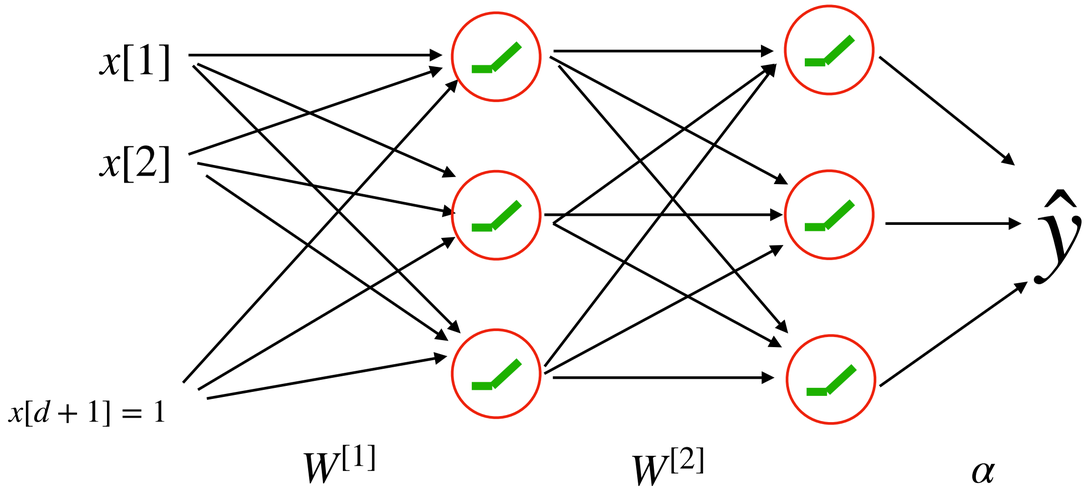

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Compute gradients:

$$\frac{\partial \ell(h(x), y)}{\partial W^{[1]}} \qquad \frac{\partial \ell(h(x), y)}{\partial W^{[2]}}$$

$$\frac{\partial \ell(h(x), y)}{\partial \alpha} \qquad \frac{\partial \ell(h(x), y)}{\partial b}$$

Training neural network via SGD

Mini-batch Stochastic gradient descent

$\theta = [W^{[1]}, W^{[2]}, \alpha, b]$

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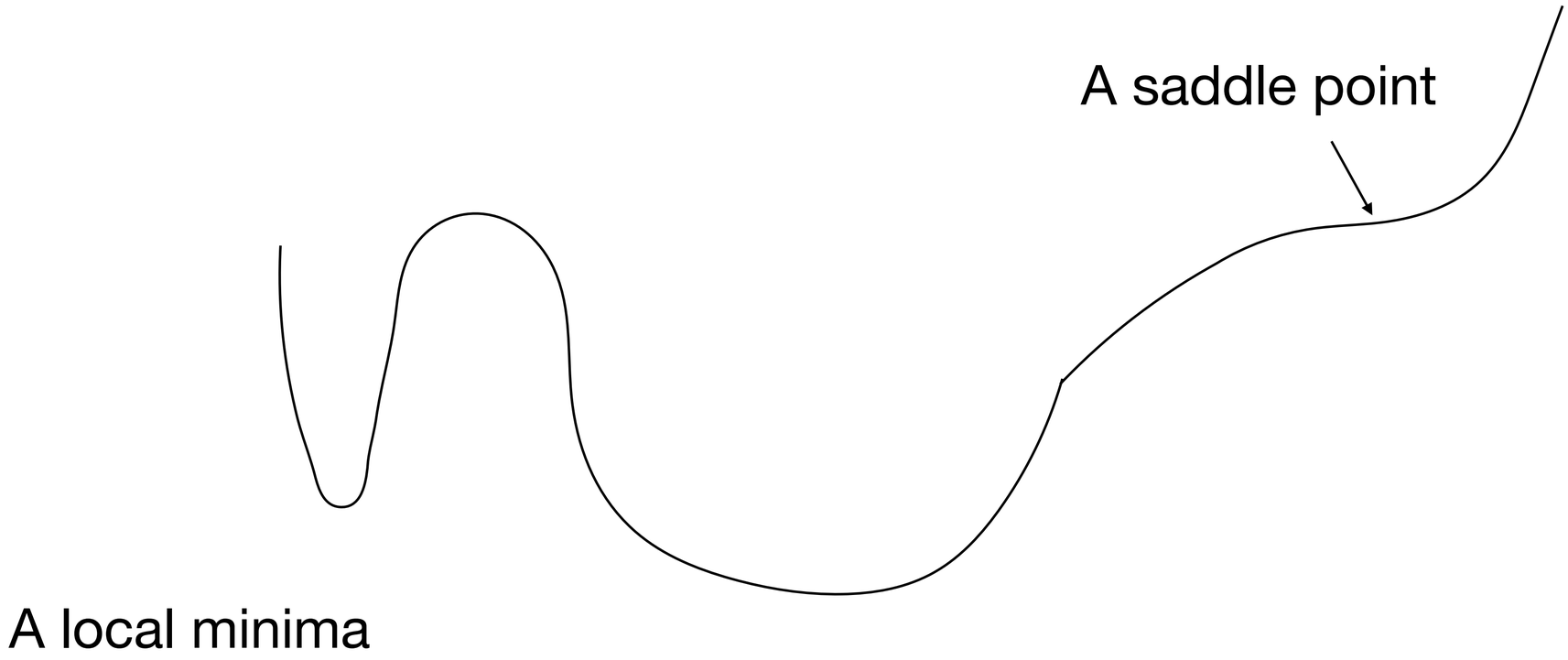
Training neural network via SGD

SGD helps avoiding local minima and saddle point



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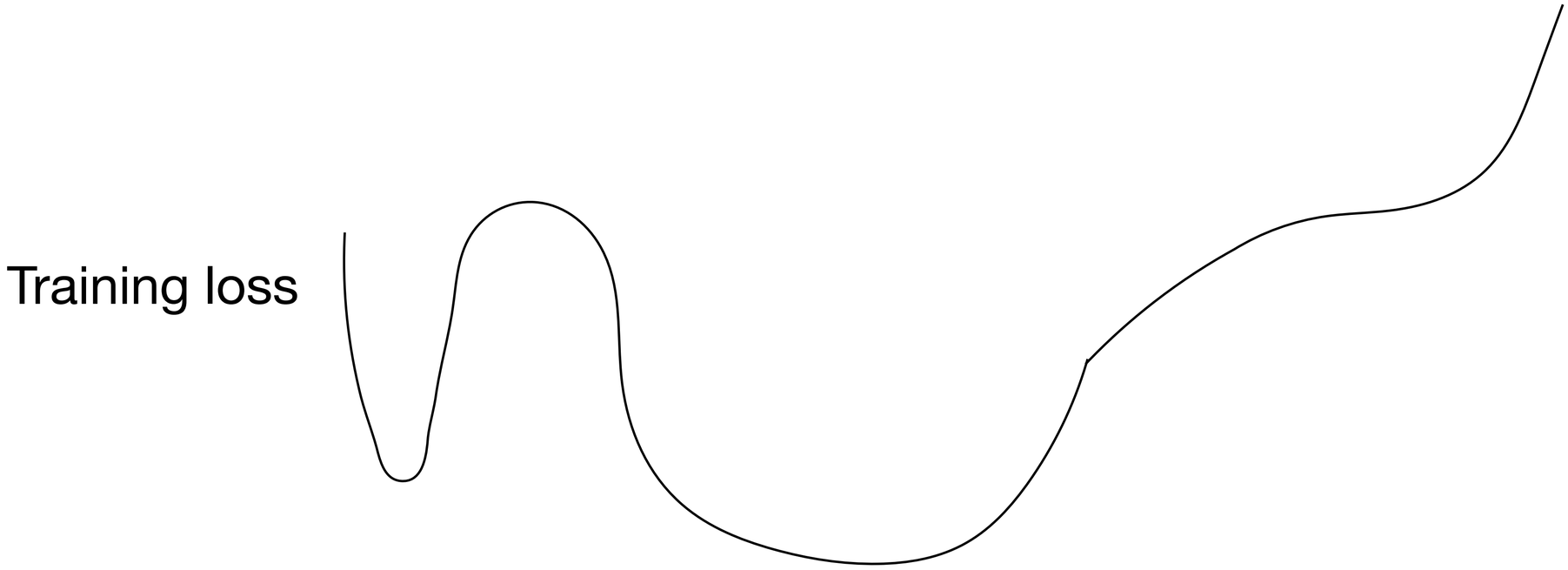
SGD tends to converge to a flat region



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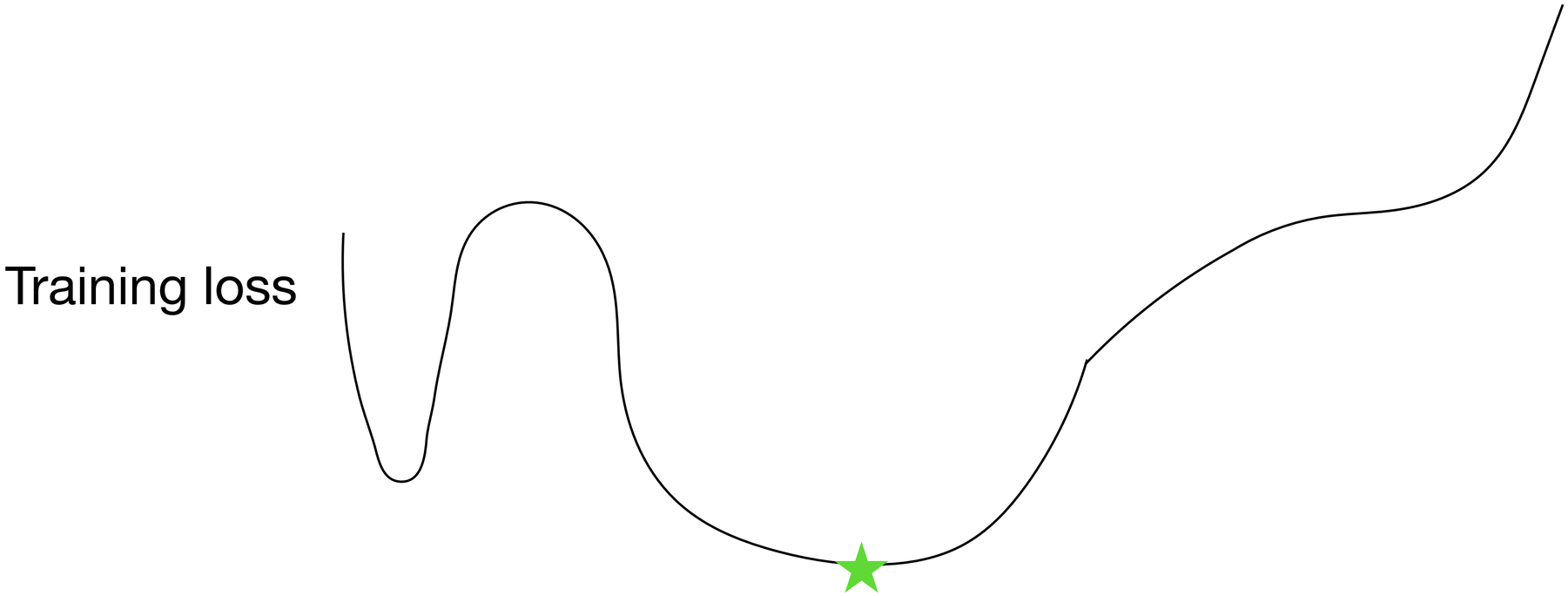

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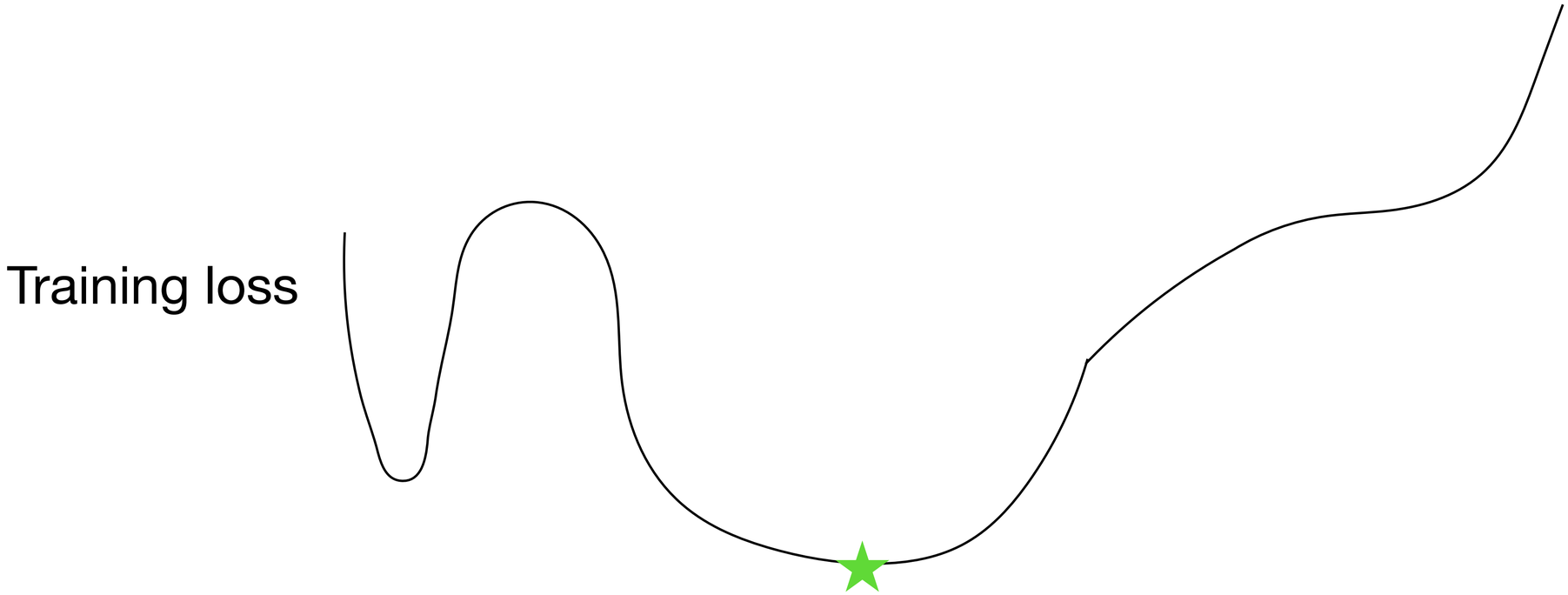

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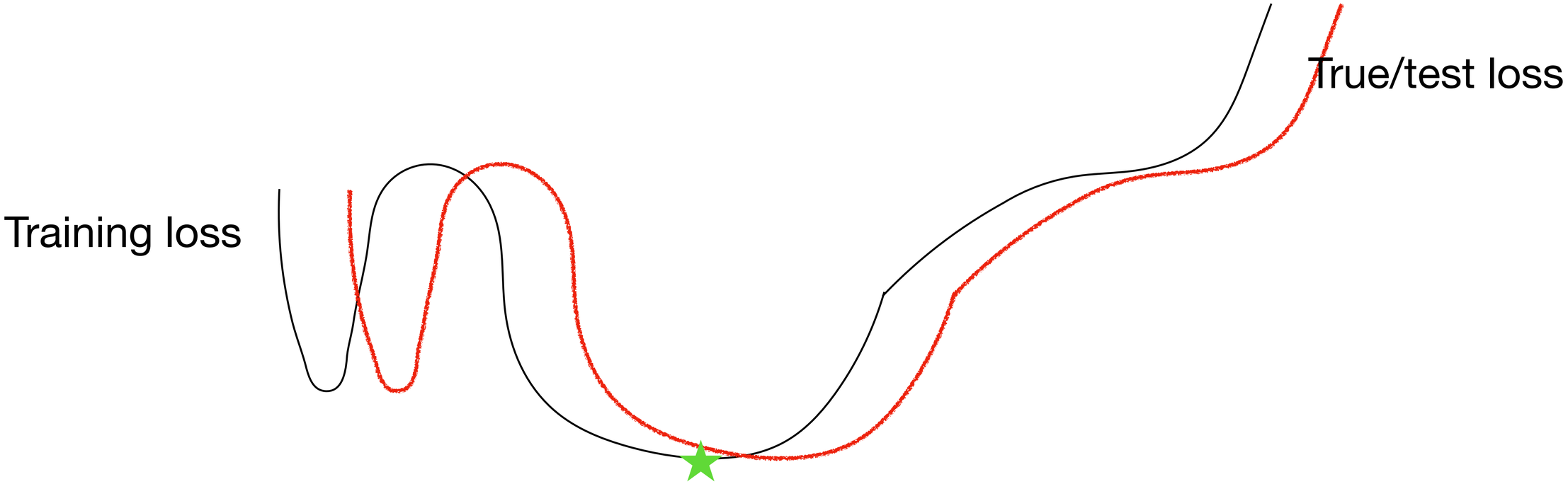

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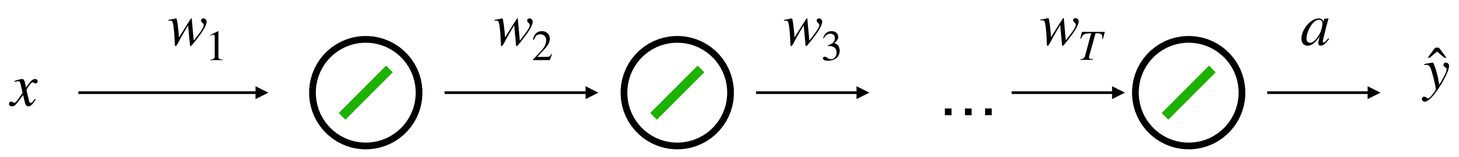

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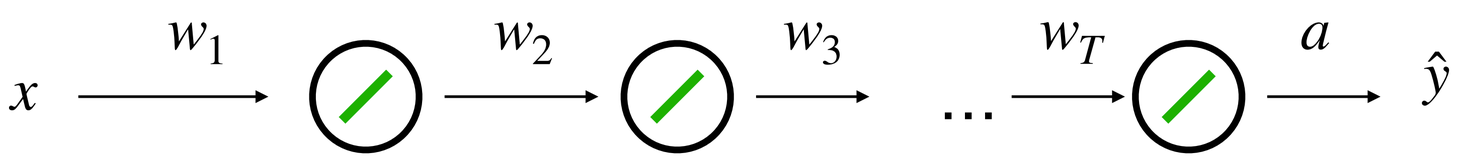

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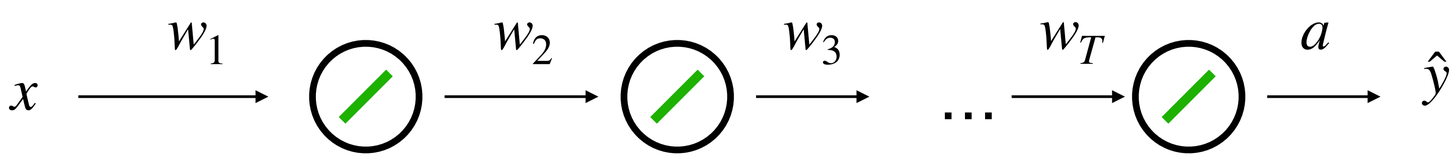

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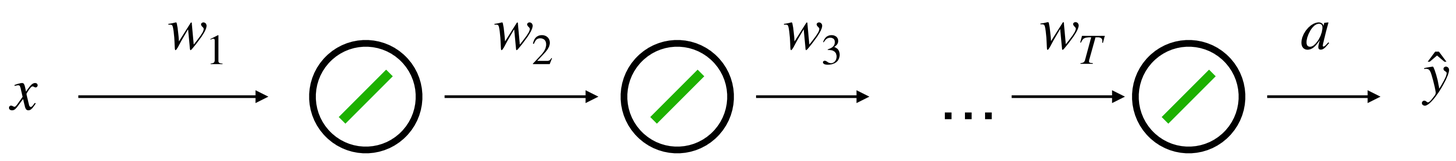

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Via chain rule: $\dfrac{\partial \ell}{\partial w_i} = \dfrac{\partial \ell}{\partial \hat{y}} \cdot \dfrac{\partial \hat{y}}{\partial w_i}$

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$$x \xrightarrow{w_1} z_1 \xrightarrow{w_2} z_2 \xrightarrow{w_3} \dots \xrightarrow{w_T} z_T \xrightarrow{a} \hat{y}$$

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Total complexity:

$$1 + 2 + \dots + T = O(T^2)$$

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Quadratic in size of the graph!

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Can we do better in running time?

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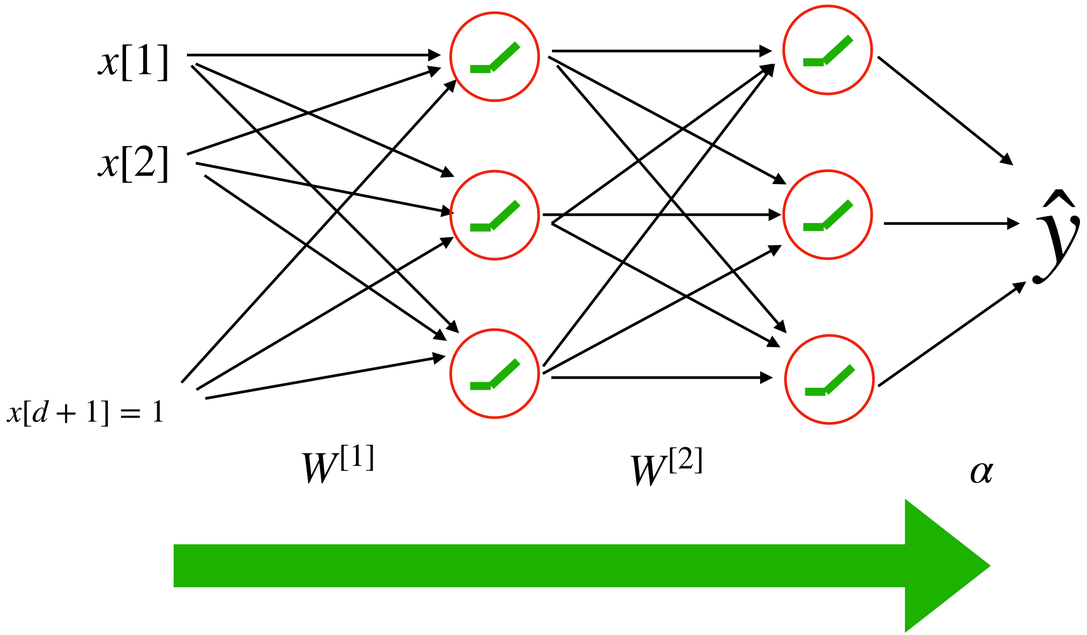
This allowed networks with intermediate "hidden" neurons between input and output layers to **learn efficiently,** overcoming the limitations noted by Minsky and Papert.

Overview of backpropagation

Forward pass followed by a backward pass

Forward pass:



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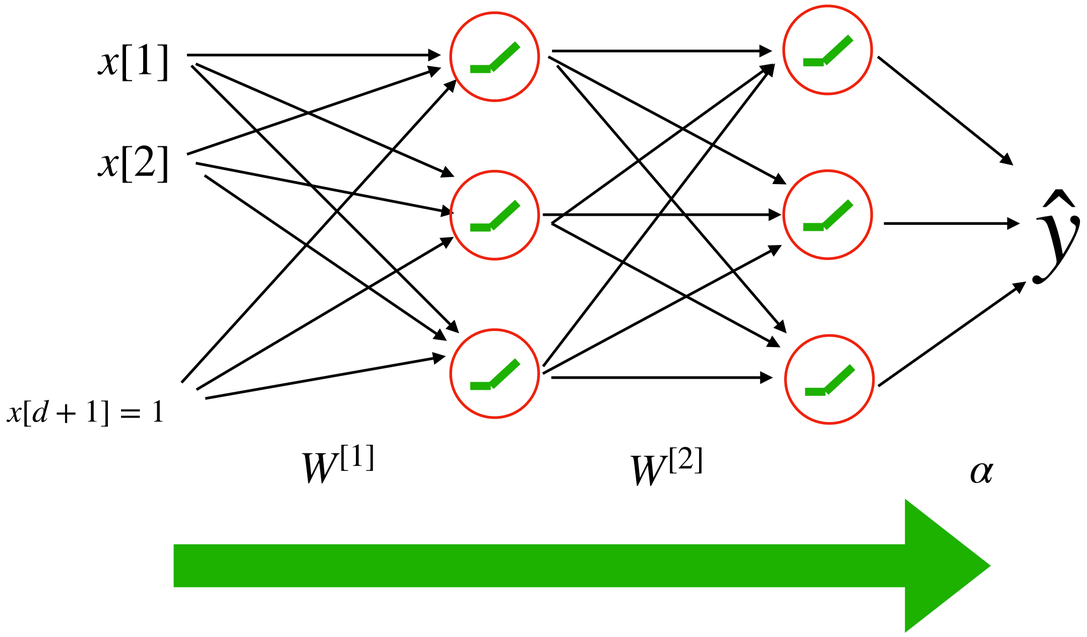

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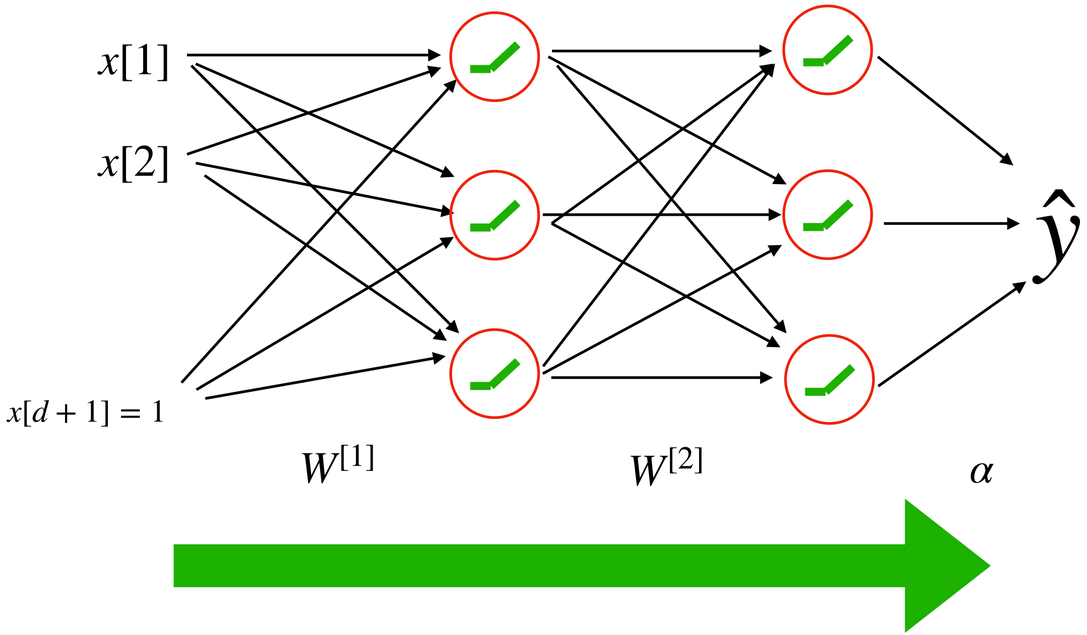

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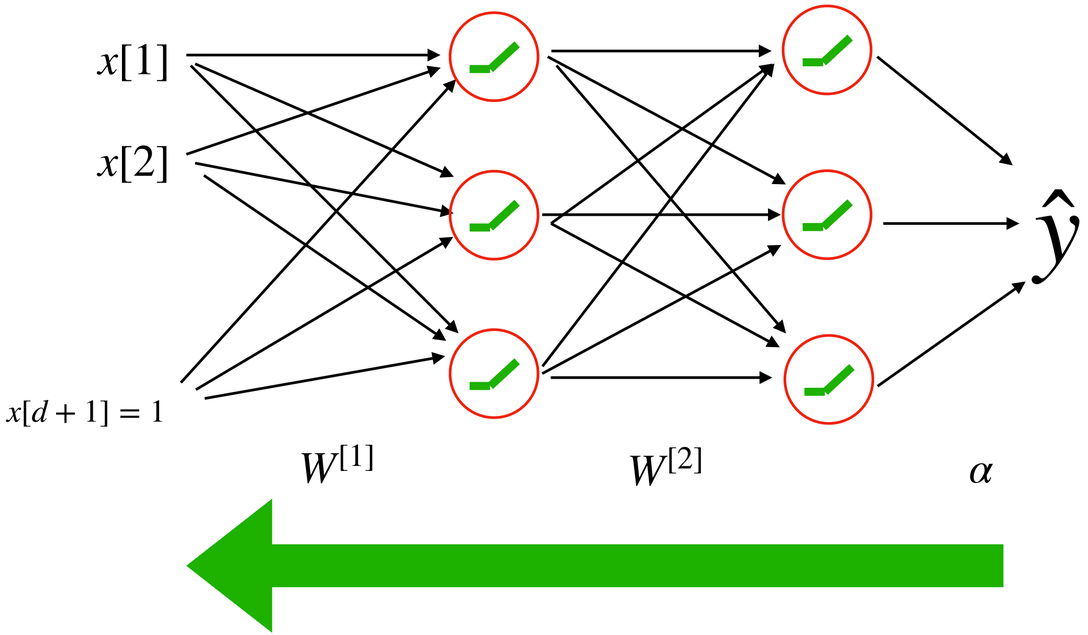

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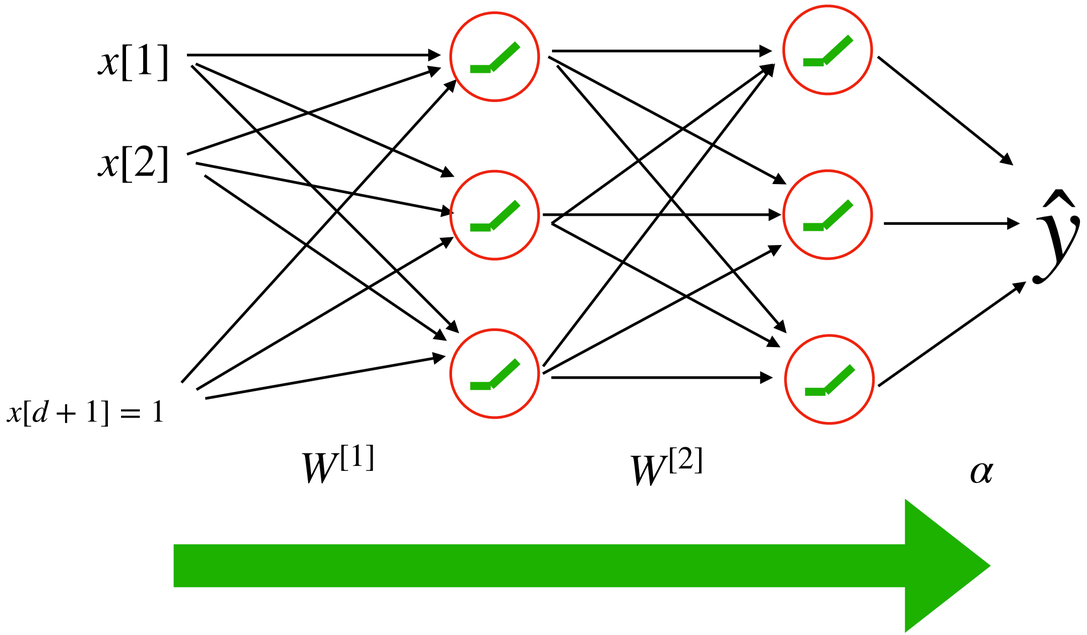

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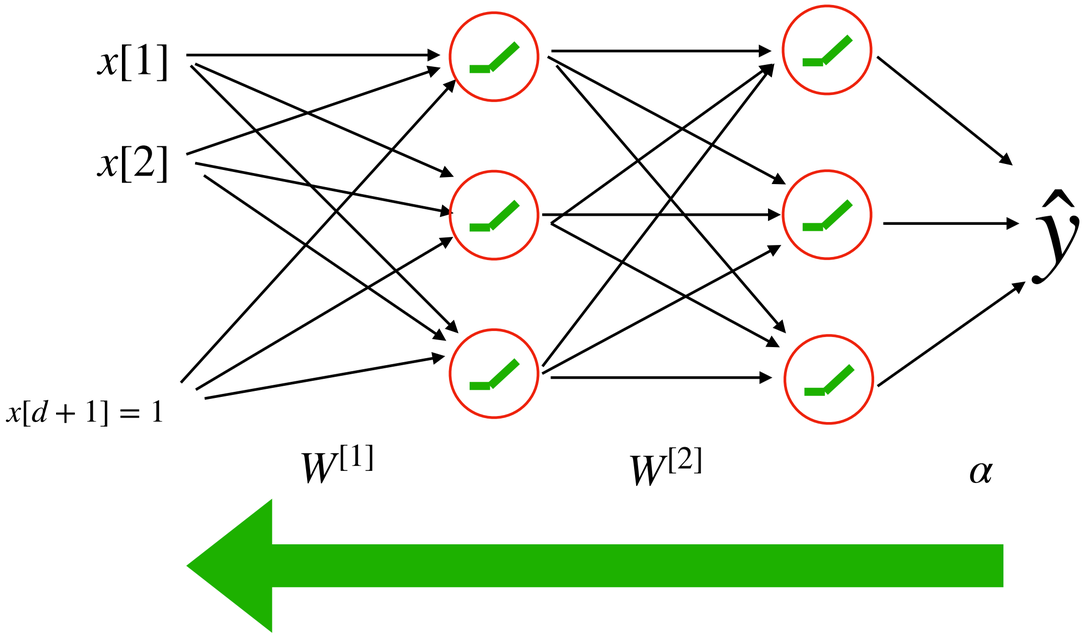
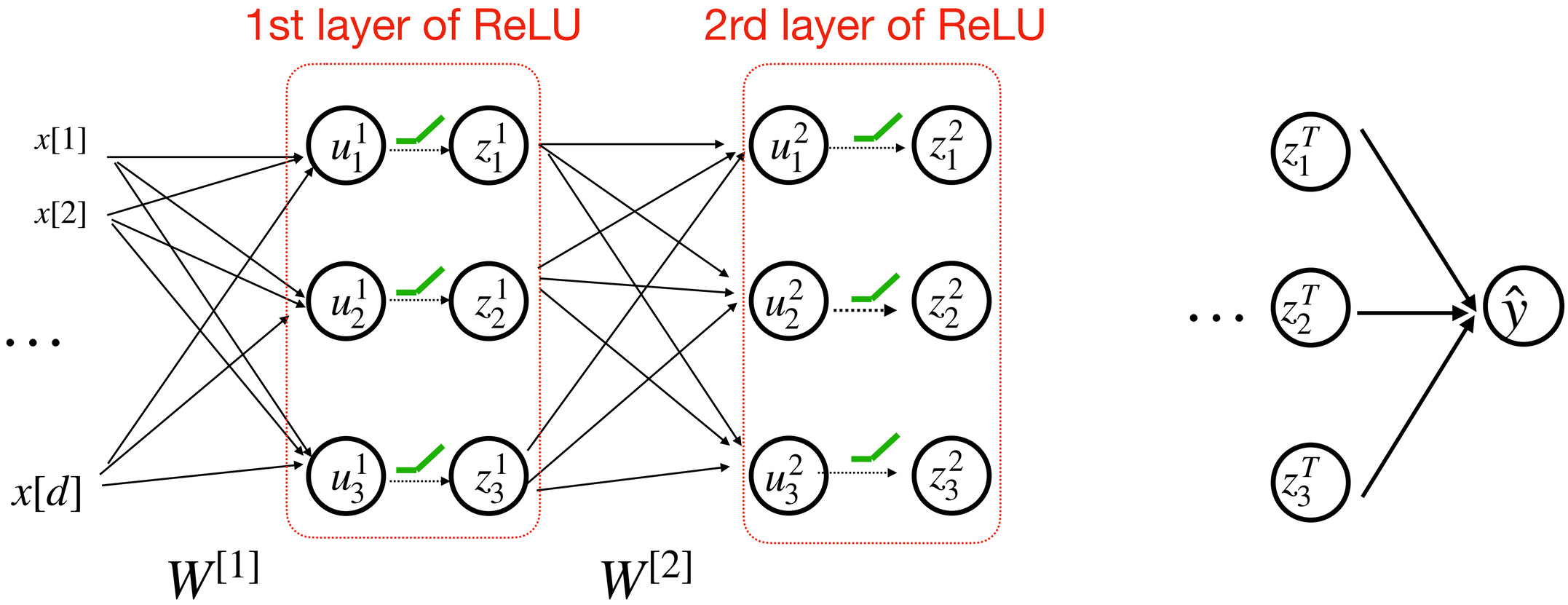


Compute derivatives

A Forward Pass: from $t = 0$ to T



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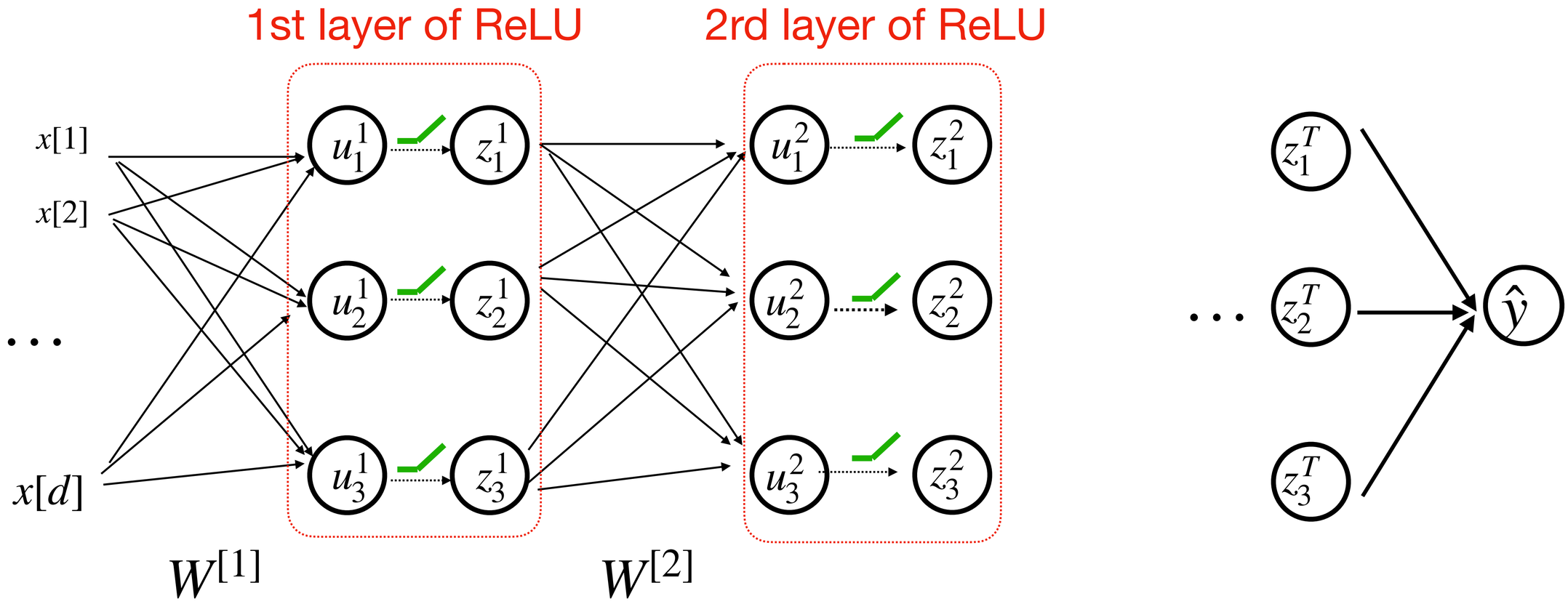

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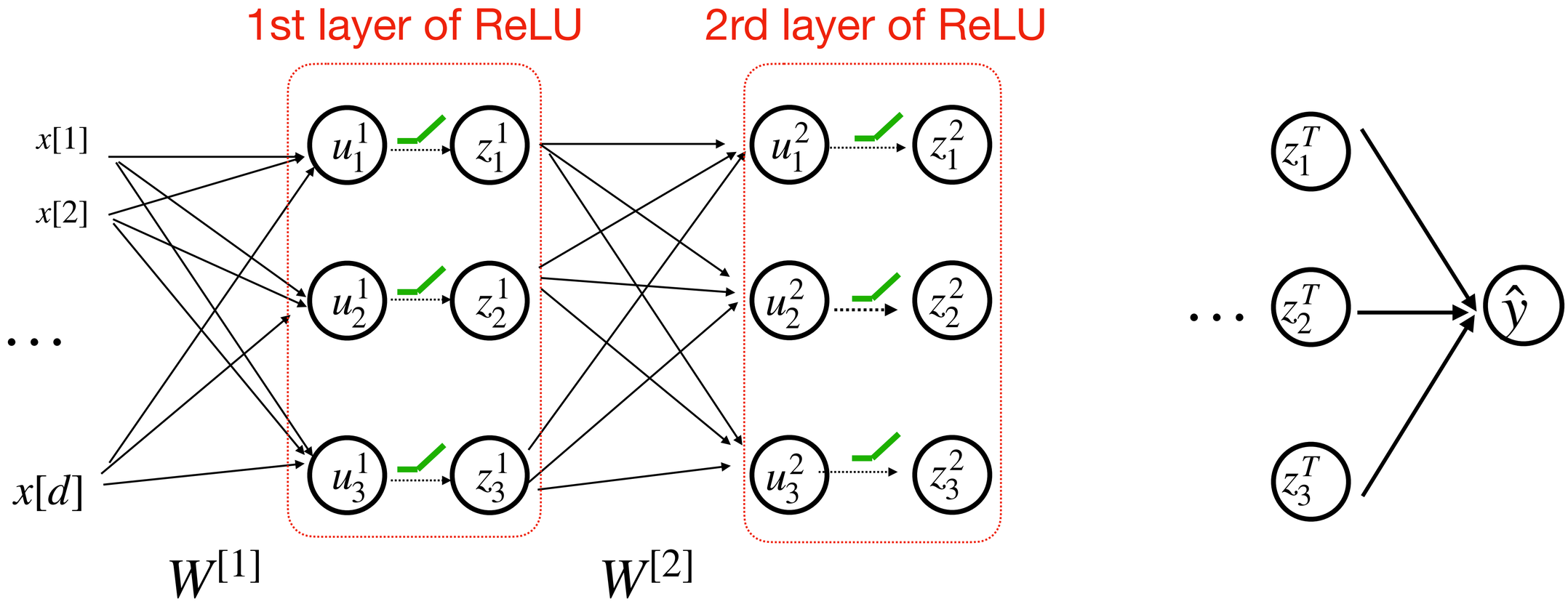
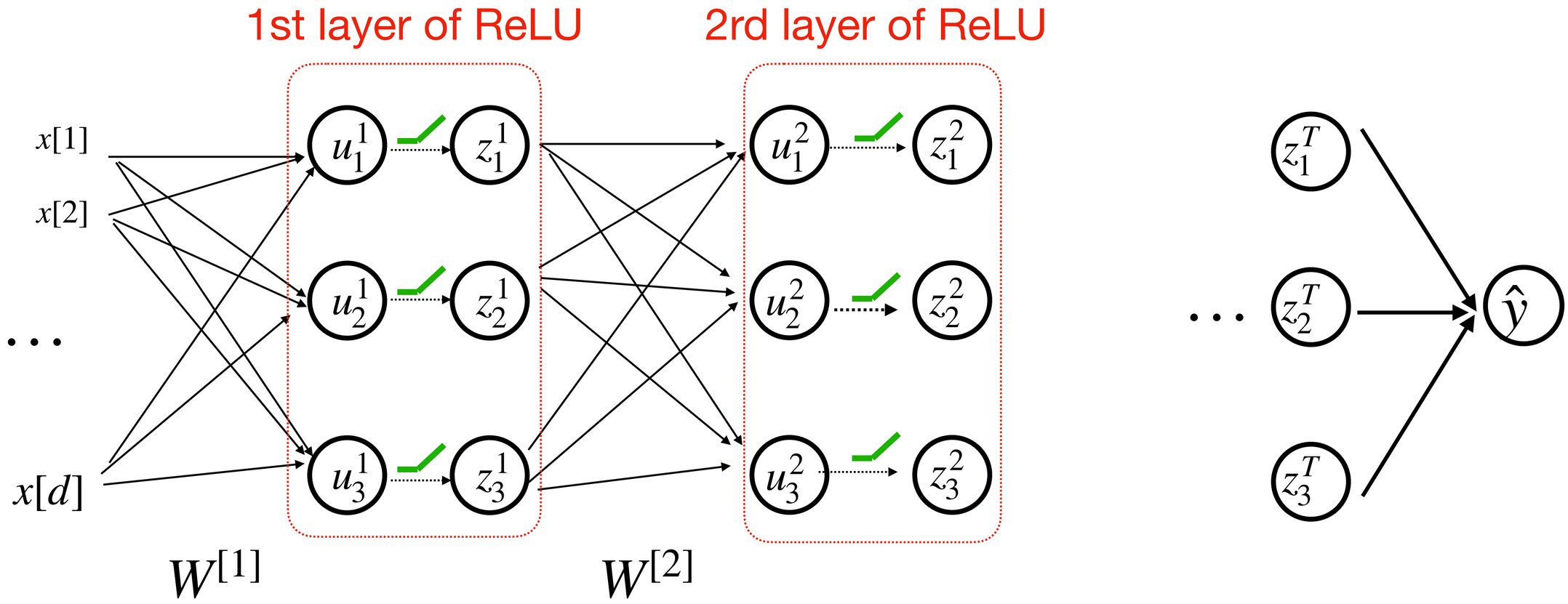


$z^0 = x$ $\quad$ $u^1 = W^{[1]} z^0$

A Forward Pass: from $t = 0$ to T



$z^0 = x$ $\quad u^1 = W^{[1]} z^0 \quad z^1 = \text{ReLU}(u^1)$

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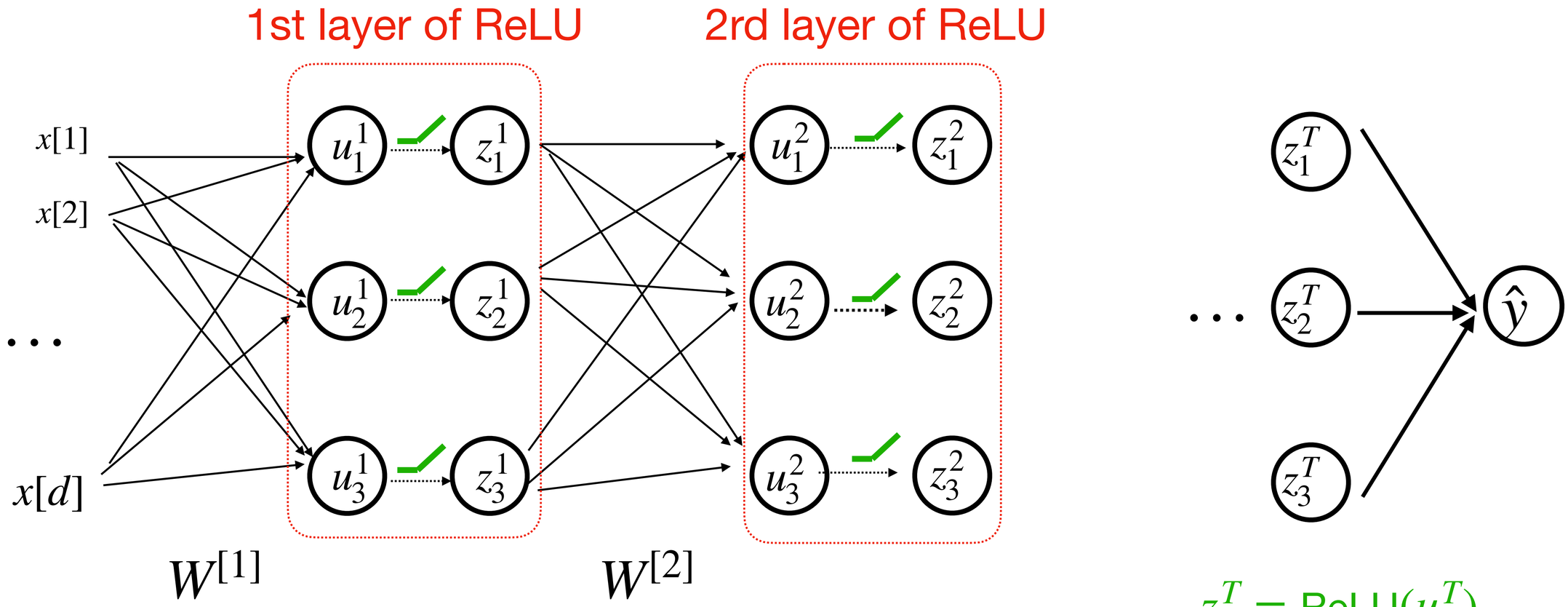
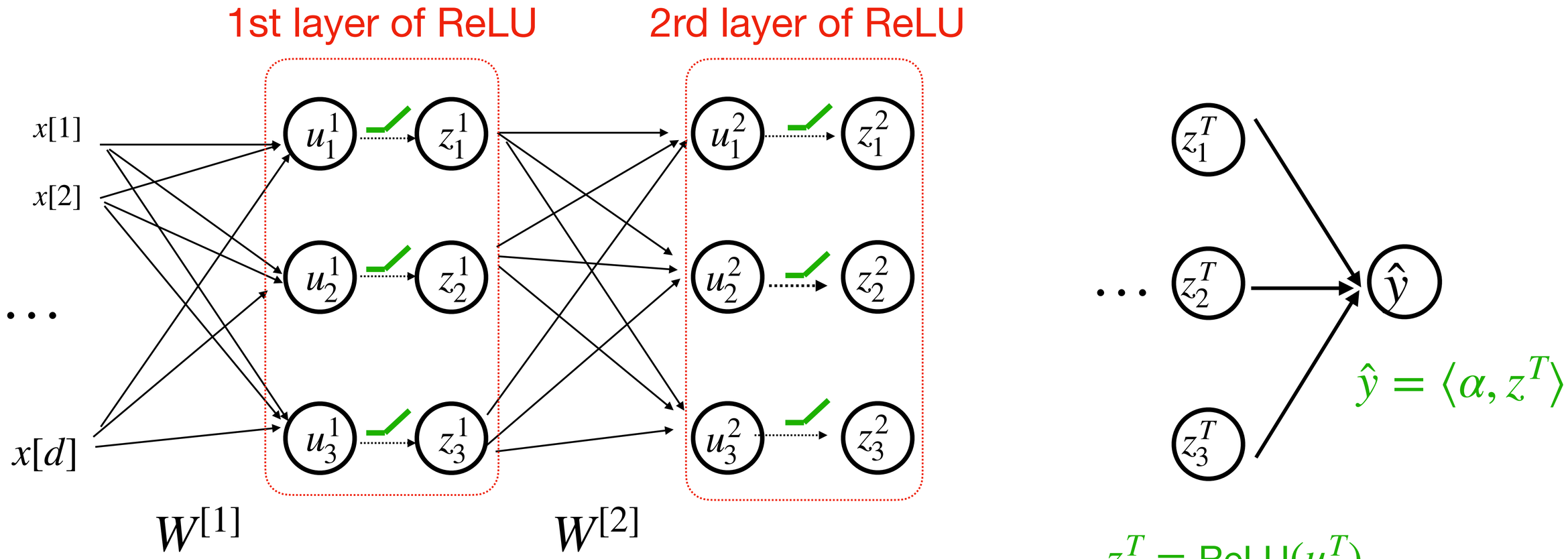


$z^0 = x$ $\quad u^1 = W^{[1]} z^0 \quad z^1 = \text{ReLU}(u^1)$

$z^T = \text{ReLU}(u^T)$

A Forward Pass: from $t = 0$ to T

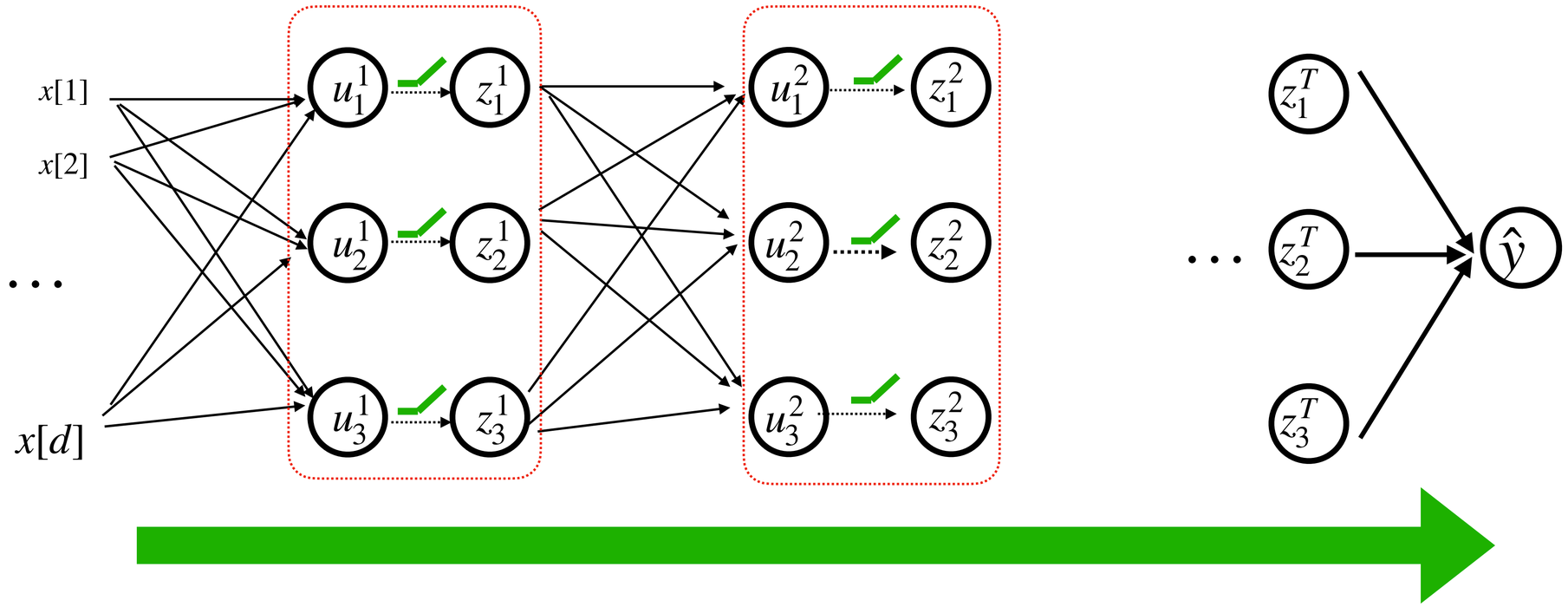


$\hat{y} = \langle \alpha, z^T \rangle$

$z^T = \text{ReLU}(u^T)$

$z^0 = x$ $\quad u^1 = W^{[1]} z^0 \quad z^1 = \text{ReLU}(u^1)$

Summary of the forward pass



All nodes' values (i.e., z, u, $\hat{y}$) are computed and stored

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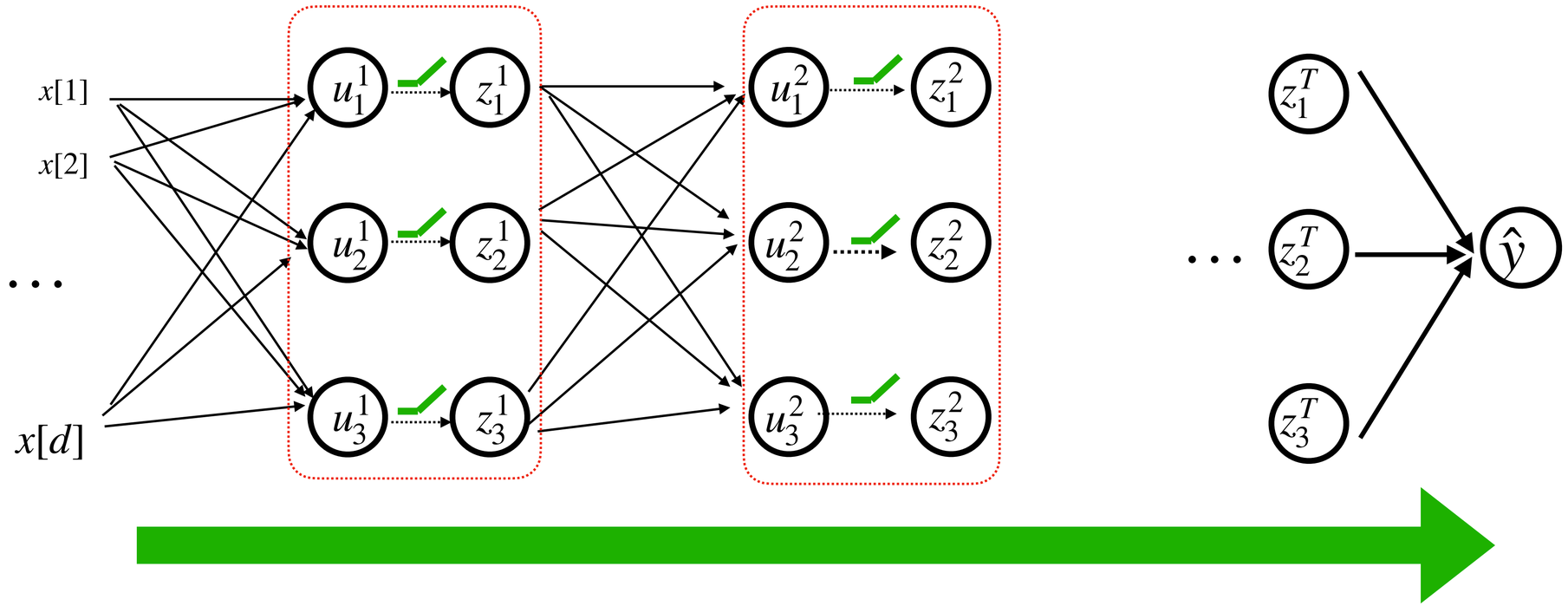

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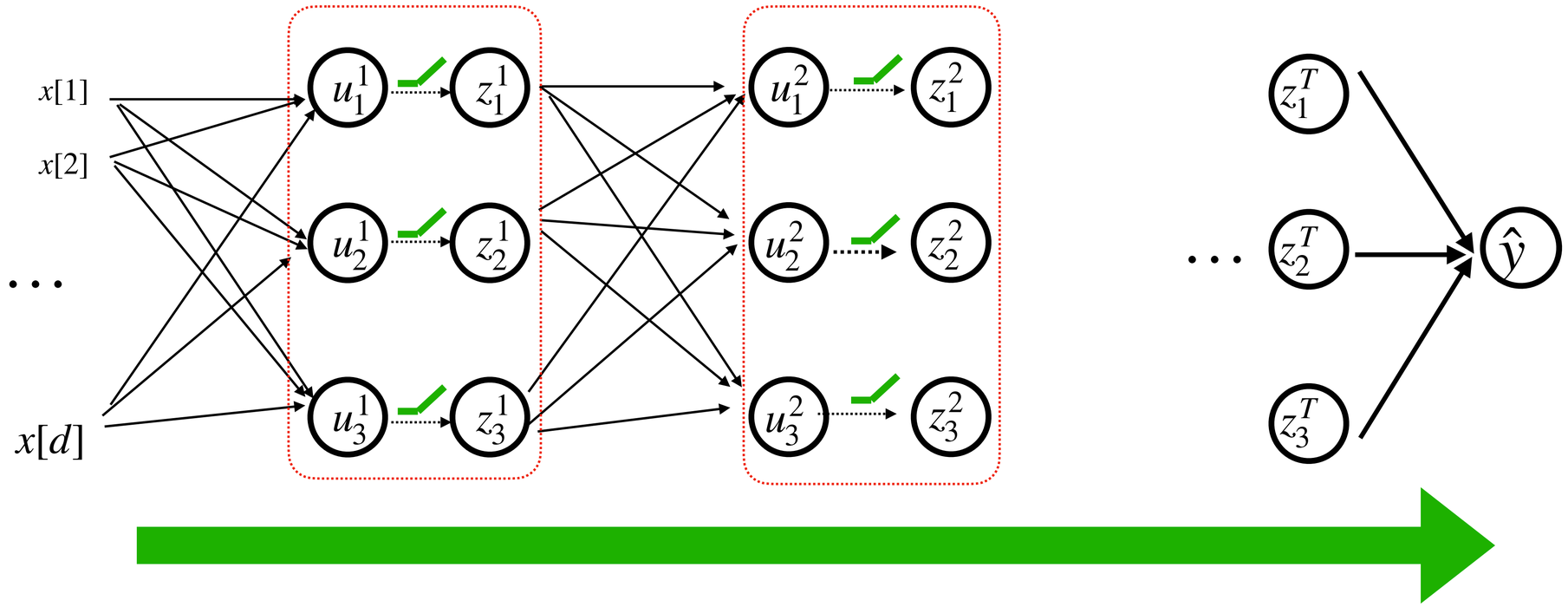

All nodes' values (i.e., z, u, $\hat{y}$) are computed and stored

Q: what is the computation complexity of the forward pass?

A: linear in # of Edges + # of nodes

The backward Pass

Claim: to compute $\partial \hat{y} / \partial w, \forall$ edge w, it suffices to compute $\partial \hat{y} / \partial z, \forall$ node z.

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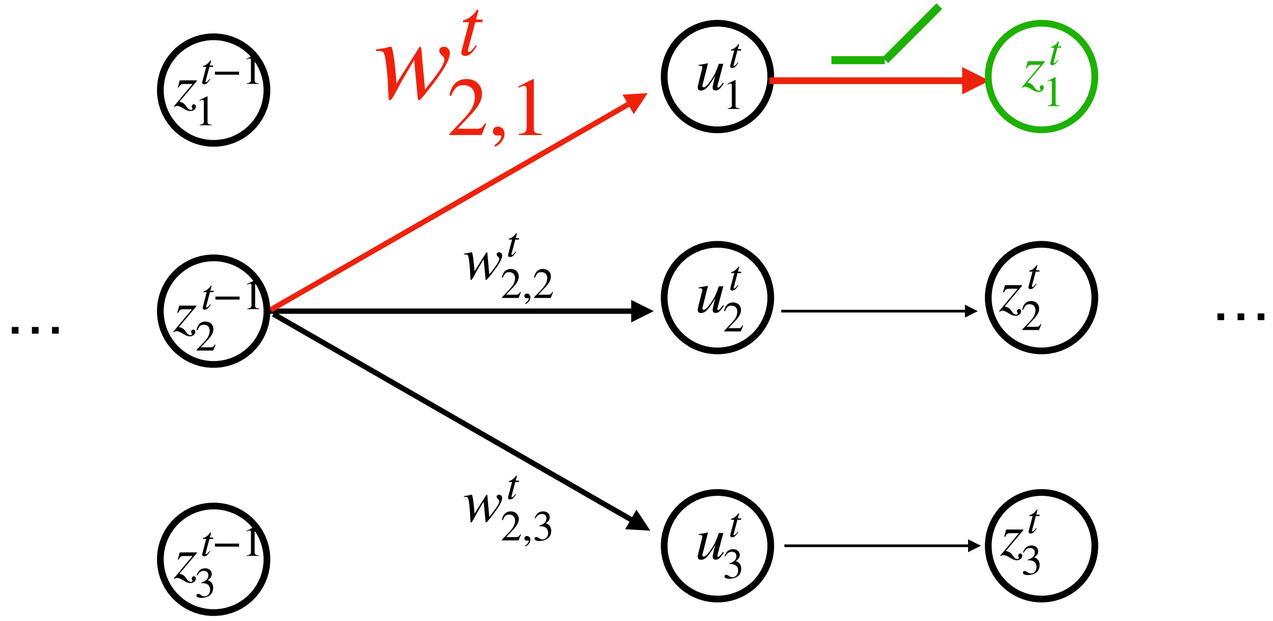

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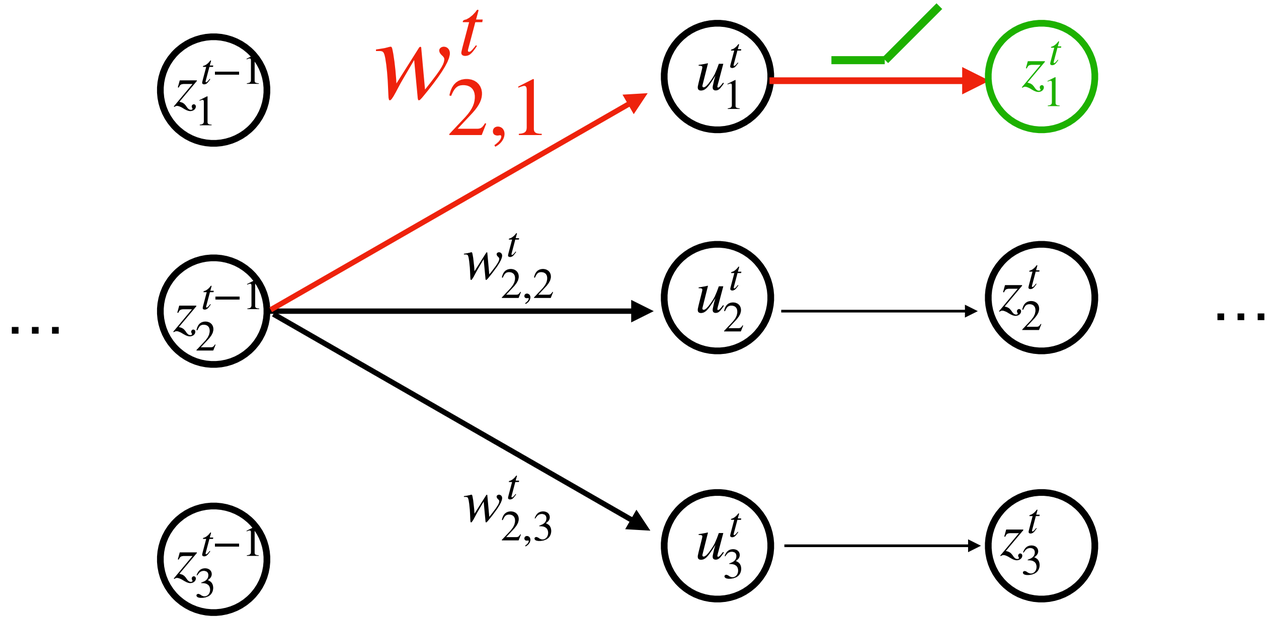

Proof:

WLOG consider $\partial\hat{y}/\partial w_{2,1}^t$

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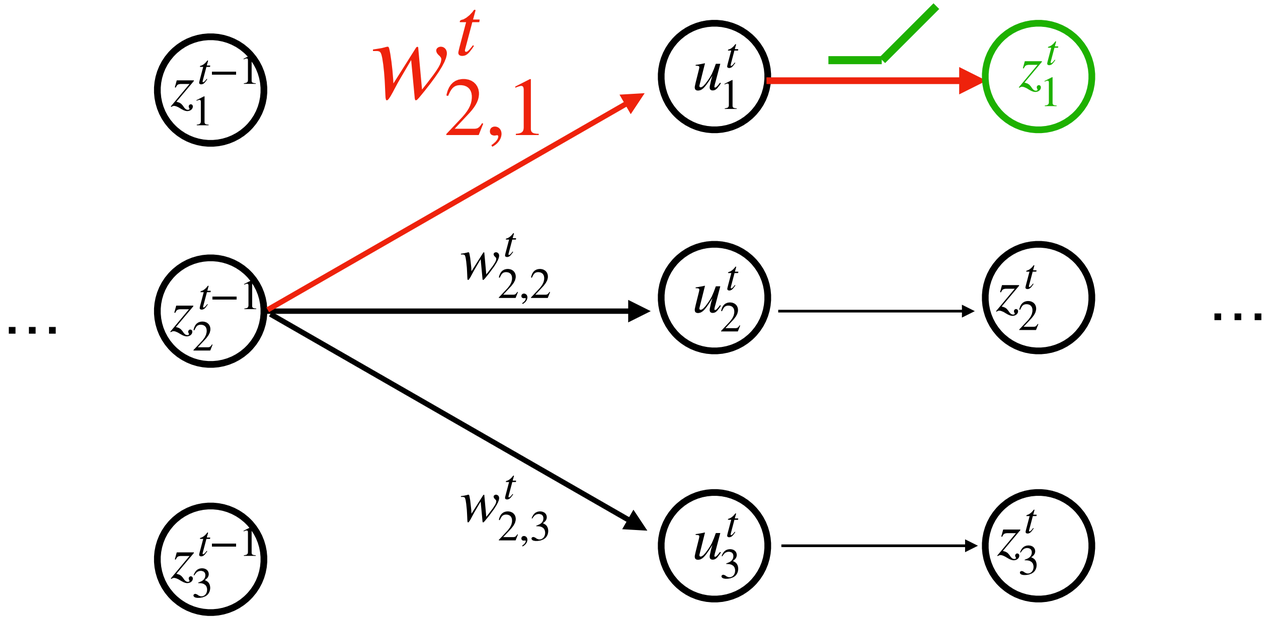

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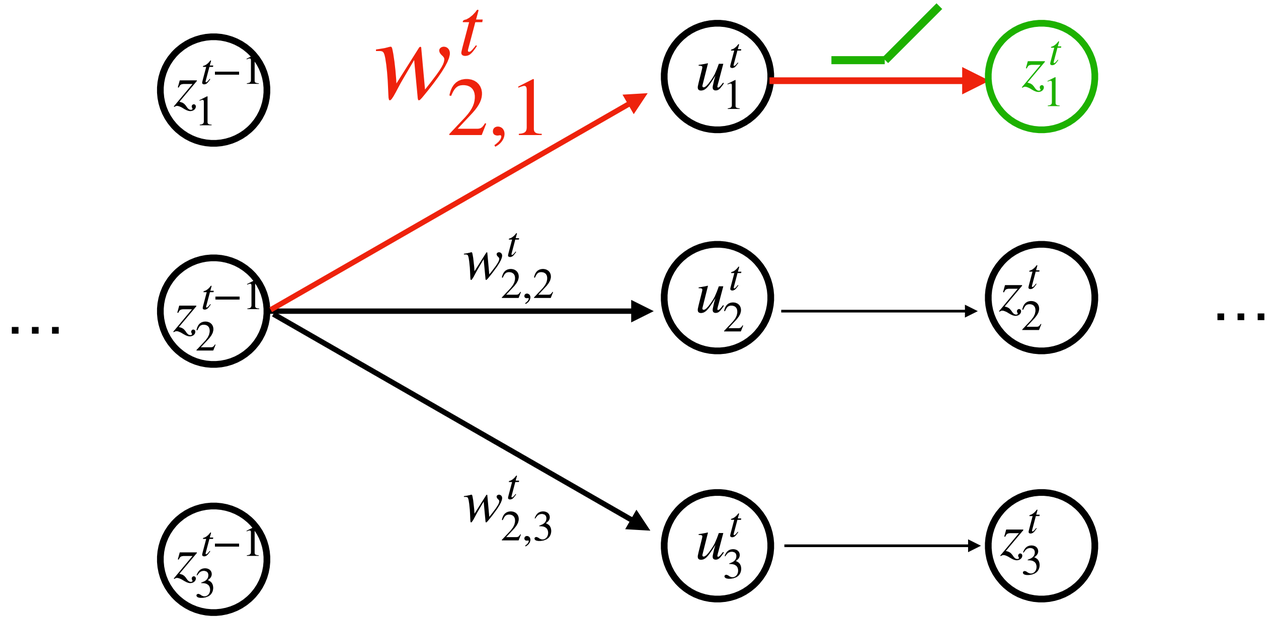

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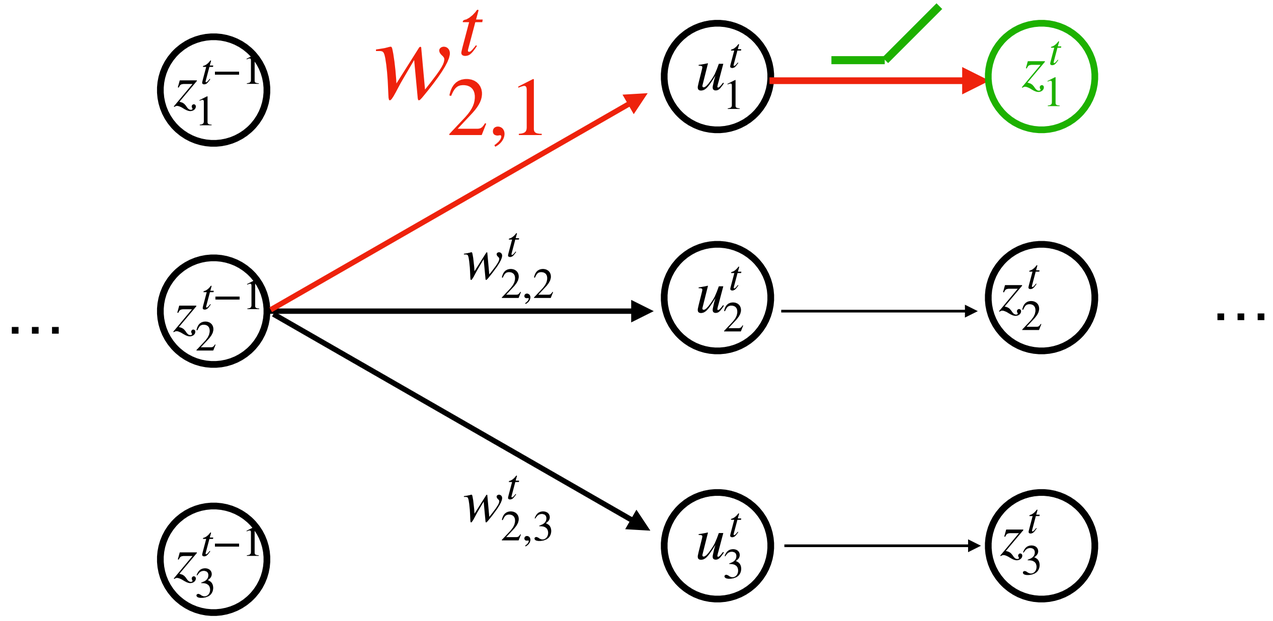

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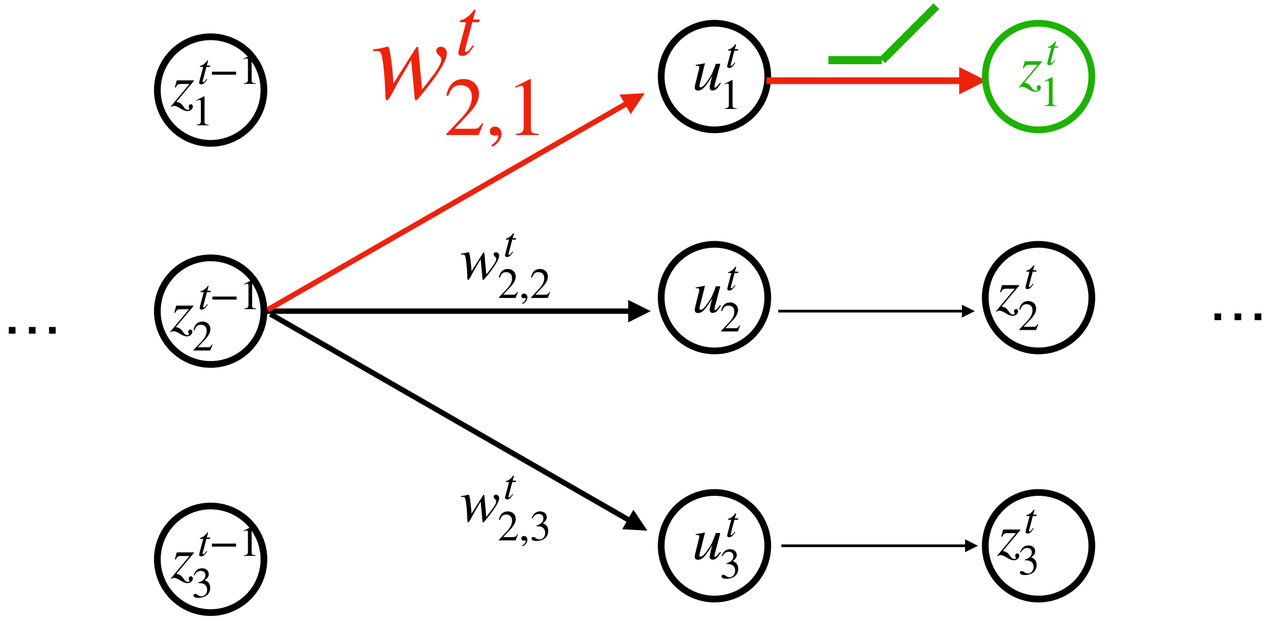

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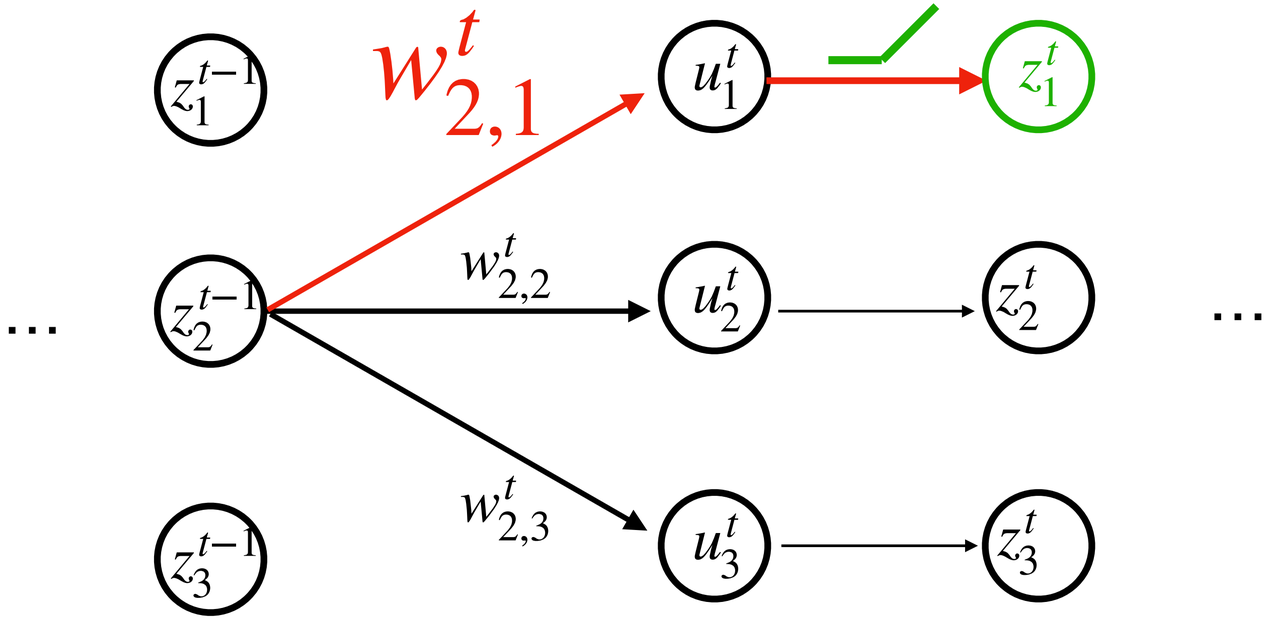

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Derivative of ReLU

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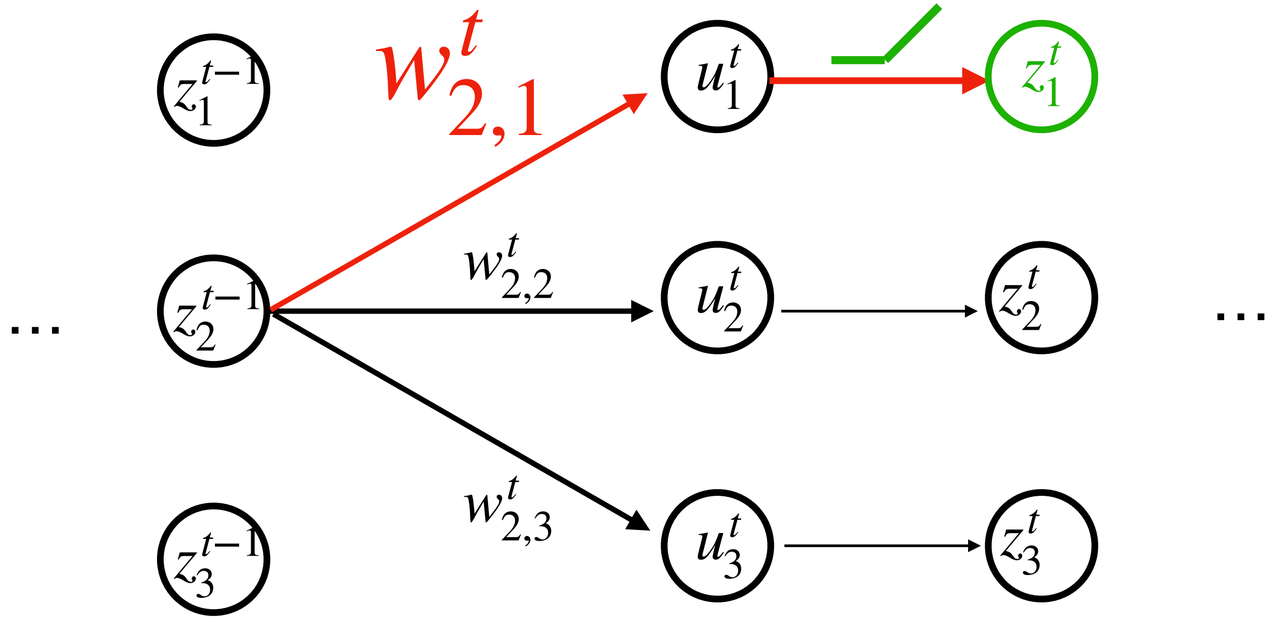

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Known from forward pass

Given by assumption

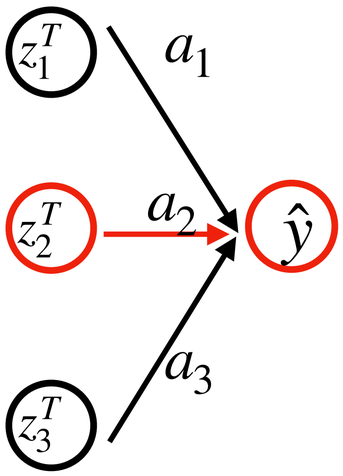
Derivative of ReLU

The backward Pass

We compute $\partial \hat{y} / \partial z^t$ backwards in time from $t = T$ to $t = 1$:

The backward Pass: base case

Base case: compute $\partial \hat{y} / \partial z^T$, for all node z at T-th Layer



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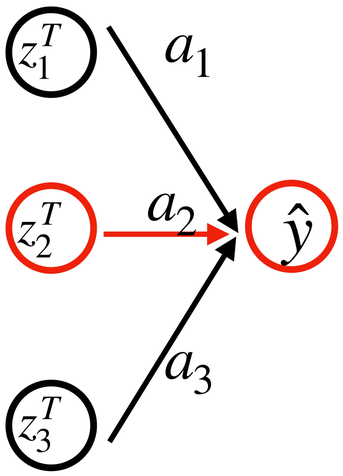
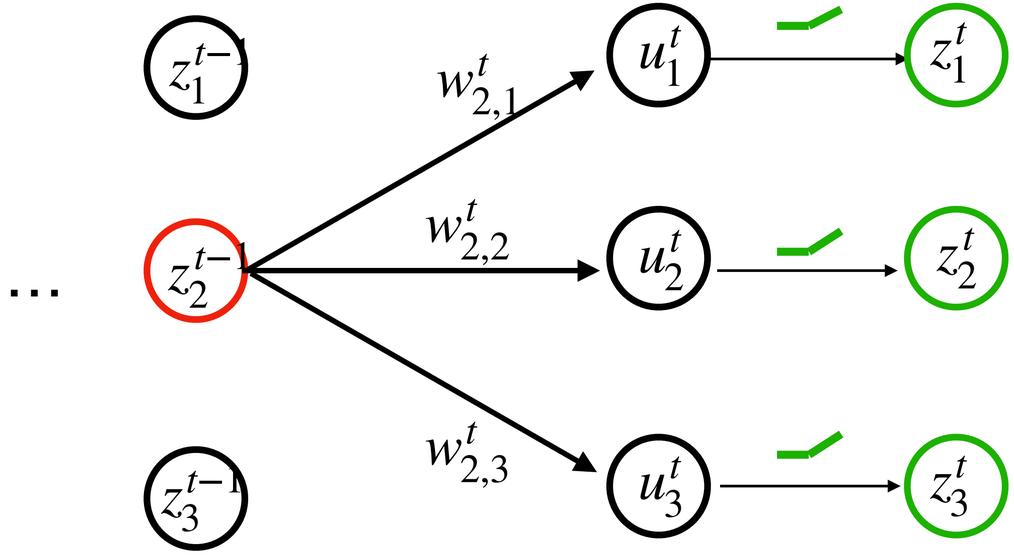


$$\partial \hat{y} / \partial z_i^T = a_i$$

The backward Pass: induction step

Assume that we have computed $\partial \hat{y} / \partial z_i^t, \forall i$



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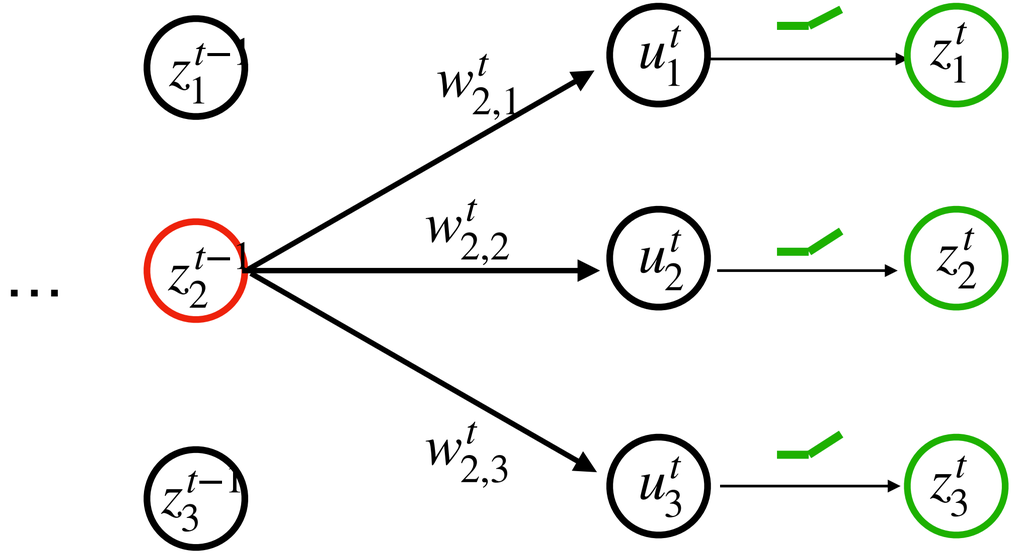

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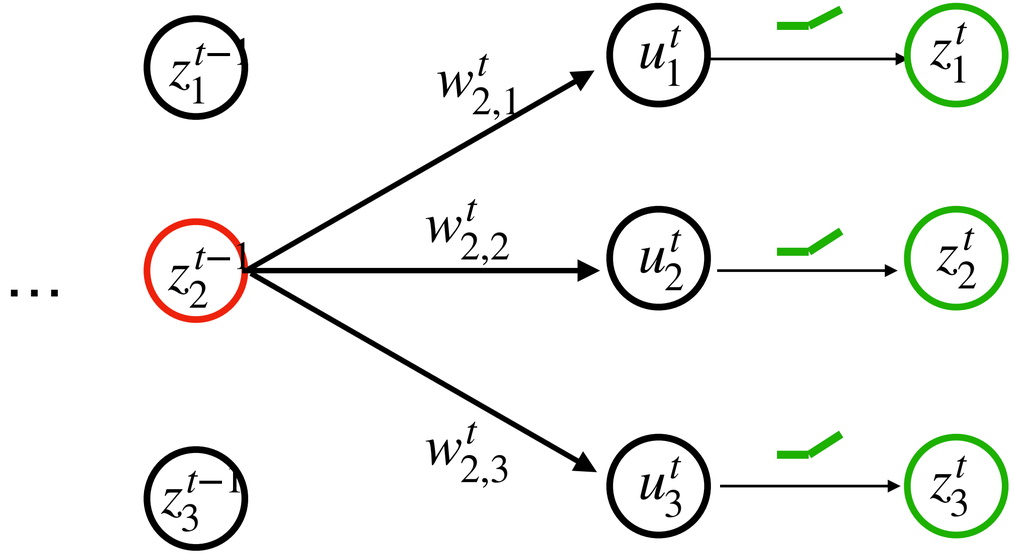

WLOG, consider $\partial \hat{y} / \partial z_2^{t-1}$

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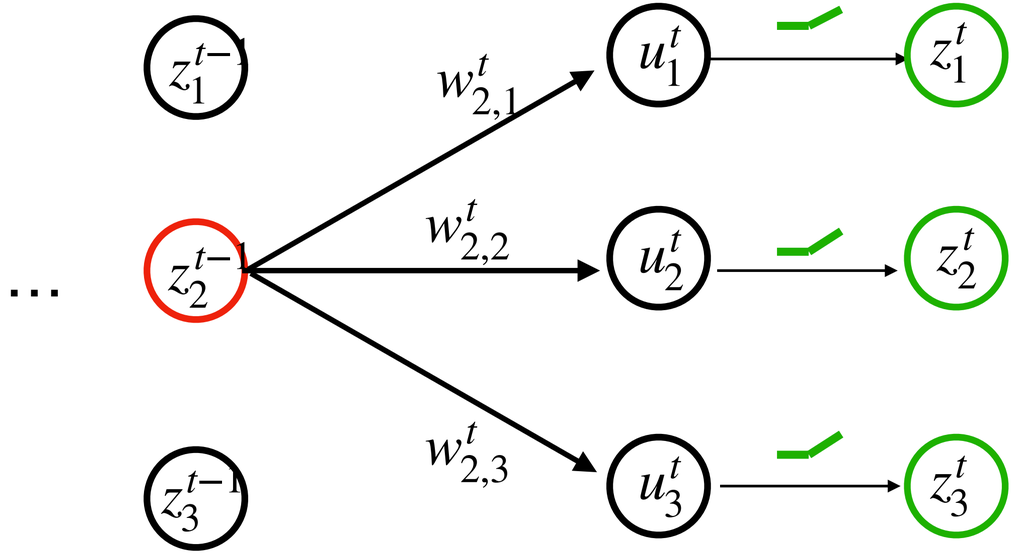

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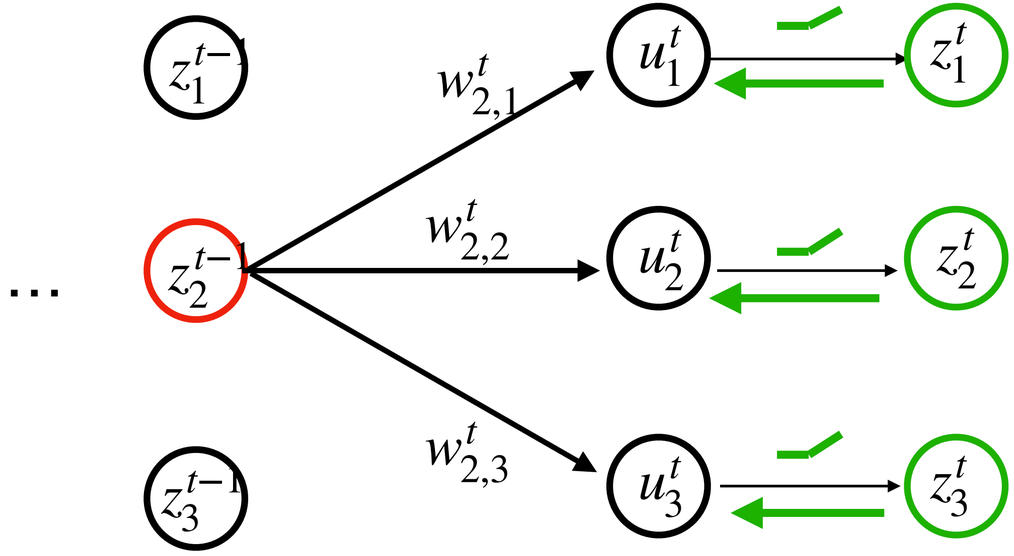

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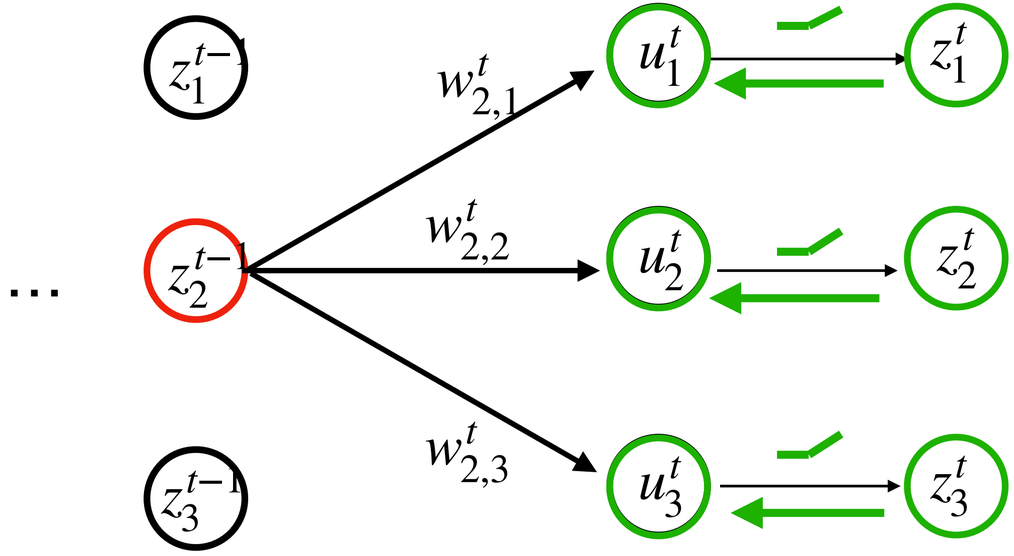

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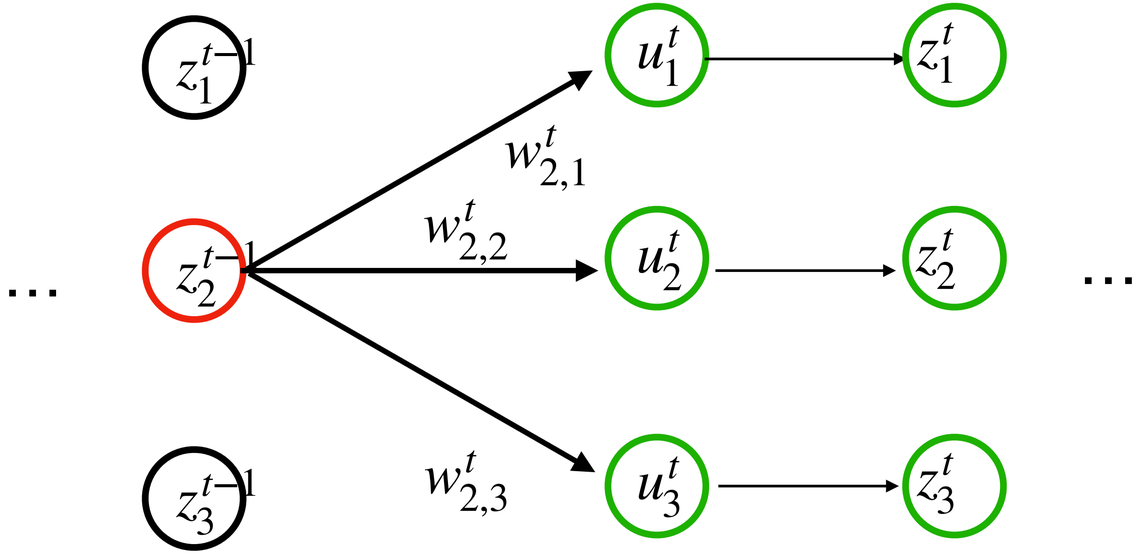
The backward Pass: induction step

Assume that we have computed $\partial \ell / \partial z_i^t, \forall i$



After step 1, we have $\partial \hat{y} / \partial u_i^t, \forall i$

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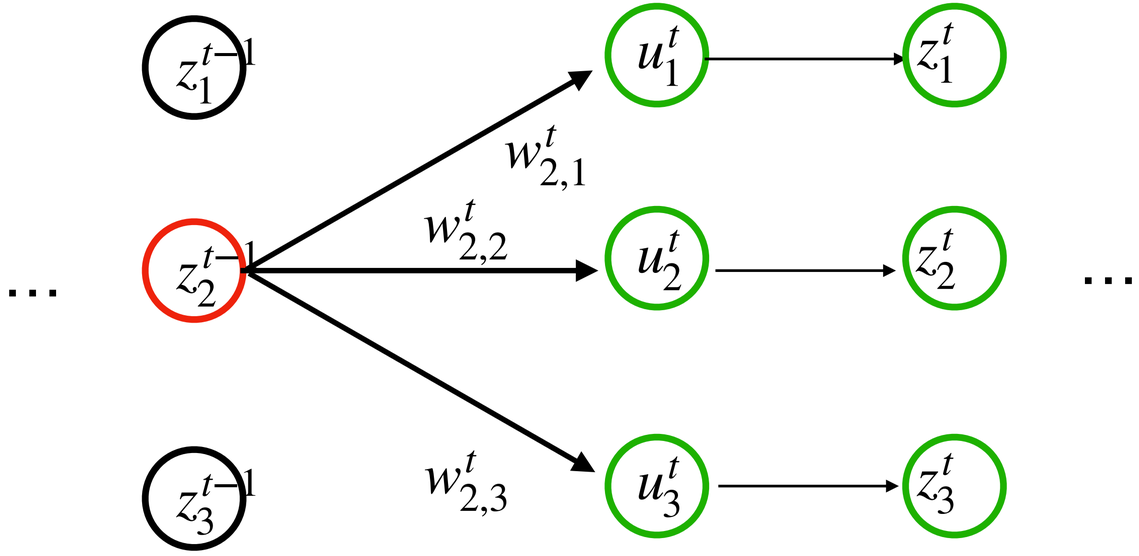

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Via multivariate chain rule:

The backward Pass: induction step

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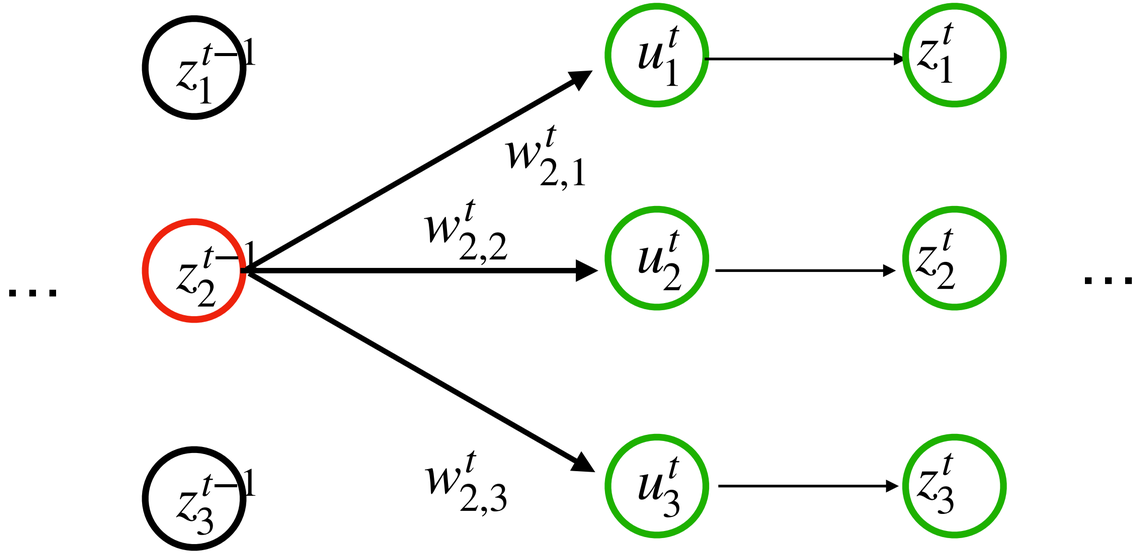

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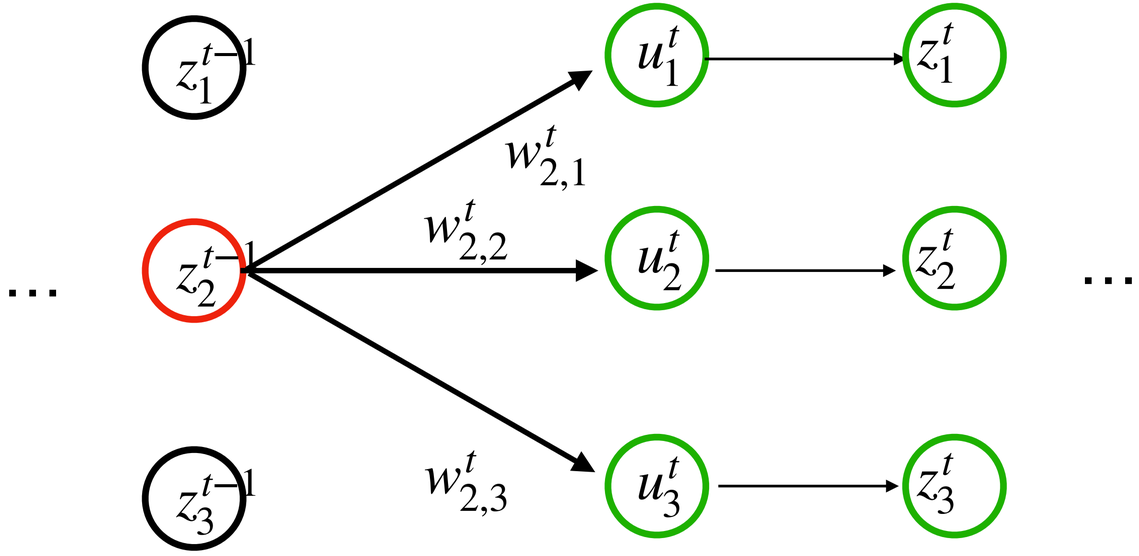

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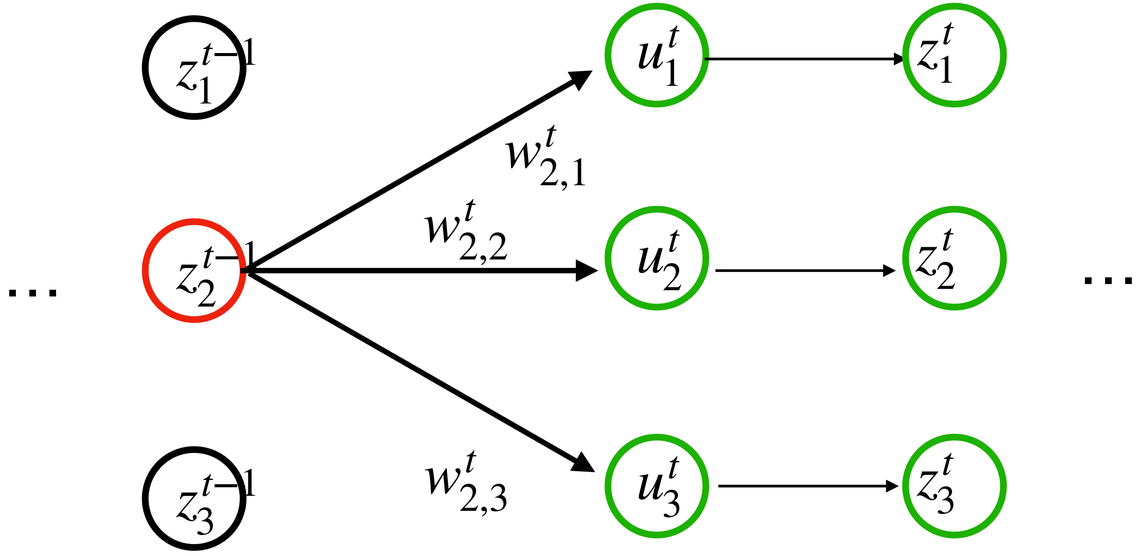

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We are done at node z_2^{t-1}!

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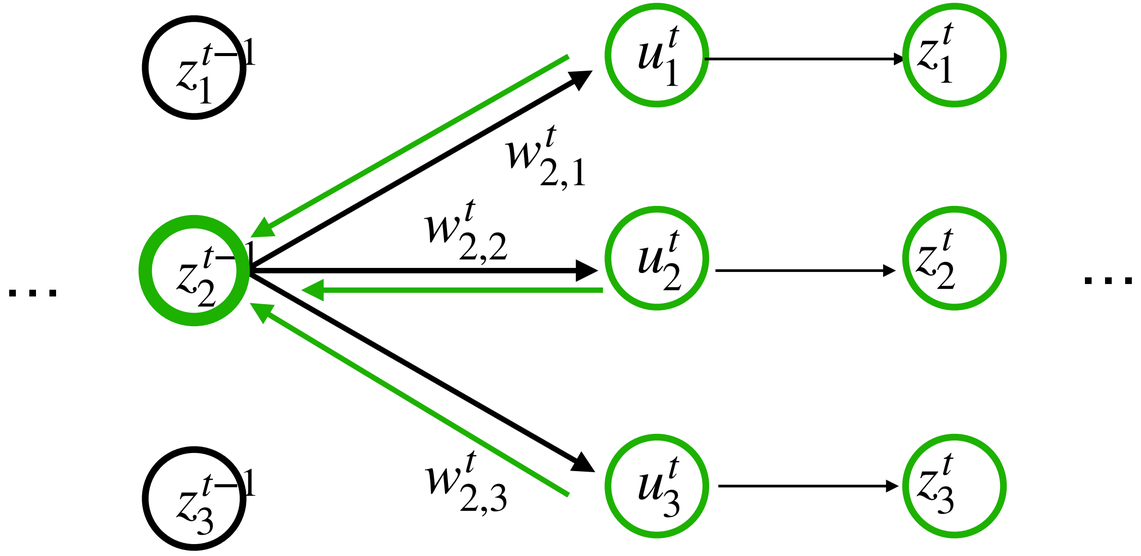

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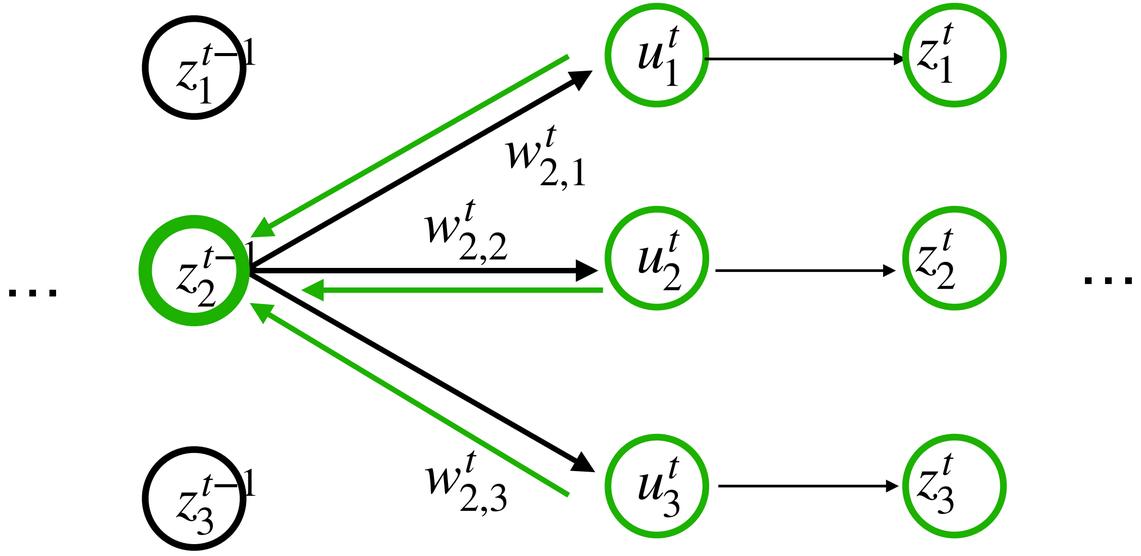

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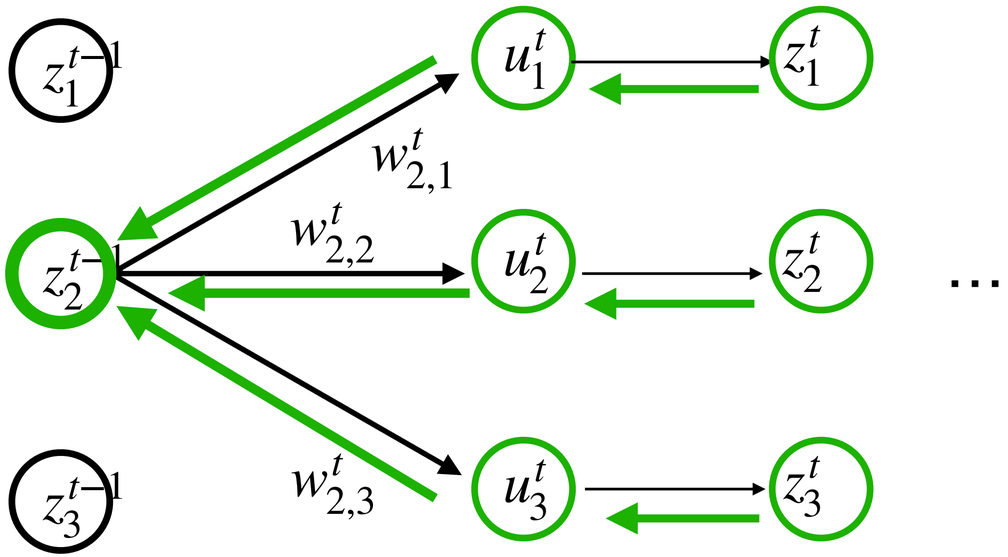
Repeat this for all $z_i^{t-1}, \forall i$

Summary of backward pass



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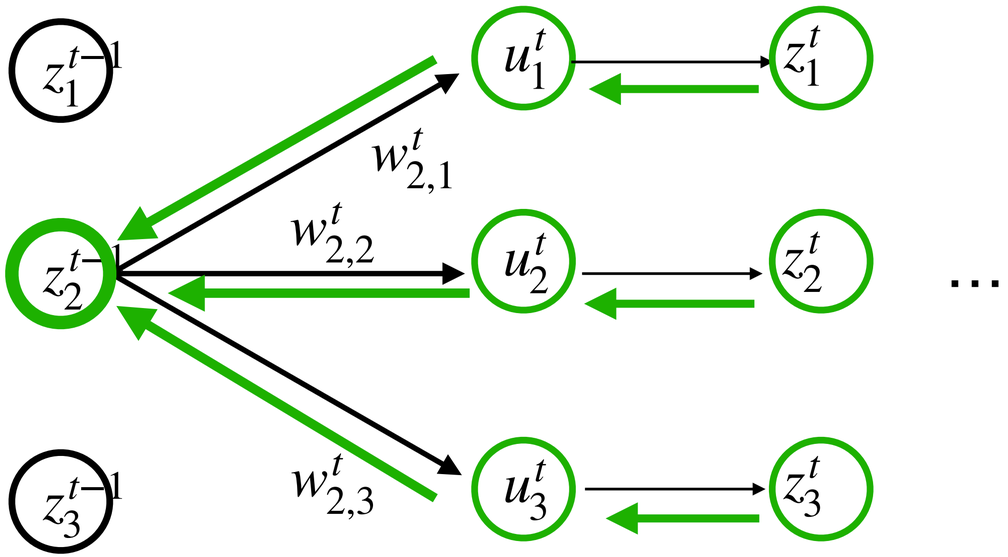

The computation from $\partial\hat{y}/z^t$ to $\partial\hat{y}/z^{t-1}$ is the # of all edges in the sub-graph

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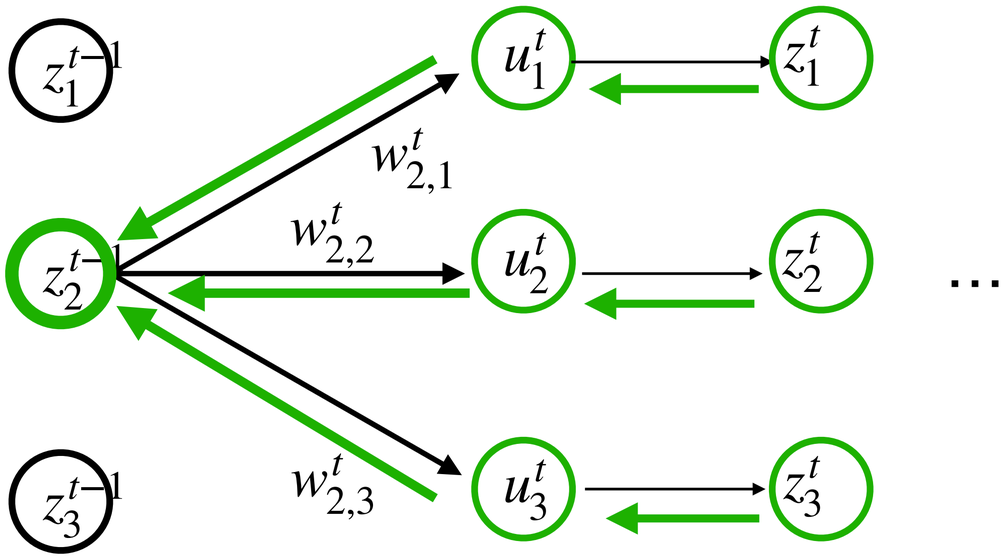

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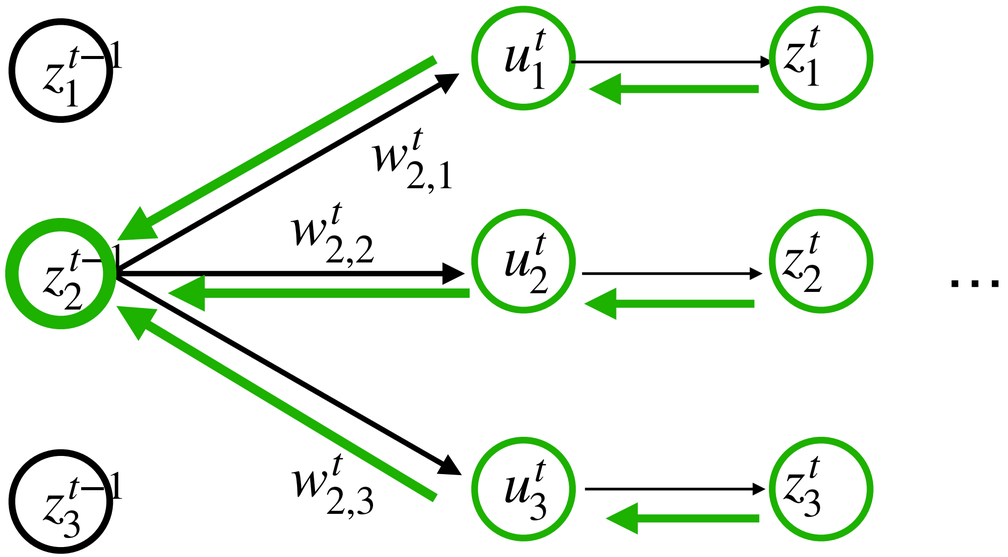
Total computation: # of edges + # of nodes!

Summary of backward pass



The computation from $\partial\hat{y}/z^t$ to $\partial\hat{y}/z^{t-1}$ is the # of all edges in the sub-graph

Total computation: # of edges + # of nodes!

Exercise: can you express backward pass in matrix-vector format?

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